# A Machine Learning Approach to Affine Conditional Moment Restriction Models

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**Abstract**

Conditional moment restriction (CMR) models provide a flexible framework for structural and reduced-form analysis. This paper develops a general framework for nonparametric estimation and inference in affine CMR models. We build on the sieve minimum distance framework of Chen and Pouzo (2015), while allowing modern machine learning methods in the first stage to accommodate flexible estimation of conditional moments with many instrumental variables. Using an orthogonalized minimum distance criterion, we characterize the requirements on first-stage learners through their prediction errors, without restricting the analysis to a particular class of learners. We provide a Wald statistic for testing smooth functionals and a Gaussian coupling of the estimated sieve coefficients, which can be used for various inference problems. We also provide lower-level verification of the first-stage requirements for selected machine learning methods.

## 1 Introduction

Conditional moment restriction (CMR) models provide a flexible framework for structural and reduced-form analysis. In a CMR model, the parameter of interest $\theta_0$ is

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identified from the conditional moment

$$m(Z, \theta_0(X)) = \mathbb{E}[\rho(O, \theta_0(X))|Z] = 0, \tag{1}$$

where $\rho$ is a residual function, $O$ is a vector of observed variables, $X$ is a subset of regressors that is an input of $\theta_0$, and $Z$ is a vector of conditioning variables or instruments. CMR models incorporate a wide range of empirical problems, including demand estimation (Blundell, Chen, and Kristensen, 2007; Compiani, 2022) and heterogeneous treatment effects (Abrevaya, Hsu, and Lieli, 2015), among many others. Particularly, it includes all instrumental variable estimators (Chen and Pouzo, 2015), which is a main focus of this paper. Their broad empirical relevance has motivated a longstanding econometric literature on nonparametric estimation and inference in CMR models, using methods such as kernel and sieve estimators (Newey and Powell, 2003; Ai and Chen, 2003; Hall and Horowitz, 2005; Carrasco, Florens, and Renault, 2007; Darolles et al., 2011).

An important subclass of CMR models arises when the residual function is affine in the parameter of interest. Despite its simple structure, affine CMR models encompass a broad range of empirically relevant models, including nonparametric instrumental variable (NPIV) models, partially linear models, and many others (Blundell, Chen, and Kristensen, 2007; Florens and Sokullu, 2017; Escanciano et al., 2021; Bennett et al., 2023). Meanwhile, modern empirical research utilizes increasingly rich datasets, and in particular, many instruments are often available for identifying the structural function in many fields of economics (Gilchrist and Sands, 2016; Bastos, Silva, and Verhoogen, 2018; Deryugina et al., 2019; Alfaro, Bloom, and Lin, 2024).

While nonparametric methods provide a flexible tool for capturing those rich information contained in modern data, a major obstacle to the empirical use of nonparametric methods is their computational challenge. It is well known that the computational burden of classical methods such as kernel and sieve estimators increases exponentially with the dimension of the variables used. Perhaps for this reason, empirical researchers often rely on parametric specifications (Bastos, Silva, and Verhoogen, 2018; Deryugina et al., 2019; Alfaro, Bloom, and Lin, 2024) or low-

dimensional summaries of a larger set of instruments (Autor, Dorn, and Hanson, 2013; Burchardi, Chaney, and Hassan, 2019) even when hundreds of instrumental variables are available. This leaves a gap between the flexibility offered by nonparametric methods and the rich information available in empirical applications.

This paper addresses this gap by utilizing machine learning methods that can handle high-dimensional variables effectively. We build on the sieve minimum-distance framework of Chen and Pouzo (2015), where the estimation step involves two stages: the first-stage conditional moment estimation and the second-stage sieve coefficient estimation. We begin by approximating the structural parameter with a linear sieve. Under the affine structure, the conditional moment then reduces to

$$m(Z, \theta_n) = \mathbb{E}[c(O)|Z] + \mathbb{E}[q_n(O)|Z]^\top \beta_n, \tag{2}$$

where $c(O)$ and $q_n(O)$ are known transformations of the observed variables and $\beta_n$ is the vector of sieve coefficients. Thus, the first-stage conditional moment estimation becomes a nonparametric conditional mean prediction problem with many instruments, where modern machine learning methods are well suited.

Our core contribution is to provide a simple but general framework for estimation and inference in affine CMR models. We characterize the first-stage requirements through prediction errors, so that the subsequent theory depends on the first-stage learner only through its predictive performance. Under these conditions, we derive an asymptotic linear representation of the estimated sieve coefficients. Building on this representation, we provide Wald inference for smooth functionals and a Gaussian coupling of the sieve coefficients that can work as a building block for various inference problems. A technical challenge is that the dimension of the first-stage prediction target grows with the sieve dimension. We therefore provide lower-level verification of the prediction-error requirements for several machine learning methods using error bounds commonly available in the machine learning literature.

Our framework combines practical advantages of minimum distance and GMM estimation. Minimum distance estimation works directly with conditional moment restrictions, instead of a set of unconditional moments implied by the conditional mo-

ment. Switching the conditional moment to unconditional ones involves the choice of instrument functions and the number of unconditional moments, which can affect GMM performance (Domínguez and Lobato, 2004; Donald, Imbens, and Newey, 2009; Bennett and Kallus, 2023). On the other hand, GMM provides a natural standard error estimator once appropriate moments are specified, whereas the conventional minimum distance method may require first-stage learner-specific variance analysis (Chen and Pouzo, 2015). By orthogonalizing the plug-in minimum distance criterion function, we obtain a first-order representation and standard error estimator whose forms are agnostic to the choice of first-stage learner. This allows empirical researchers to flexibly choose machine learning methods best-suited for their applications at hand.

Our framework also provides a computationally attractive estimator by avoiding iterative training of the nested first-stage problem. In CMR models, the conditional moment is a function of the structural parameter, so evaluating the second-stage criterion function at a new parameter point during optimization may require iterative training of the first-stage conditional moment. This can be avoided by utilizing a fixed object that can be reused throughout the second-stage optimization, such as sieve space (Chen, Chen, and Tamer, 2023; Chen, Liao, and Wang, 2025), conditional mean operator (Darolles et al., 2011), conditional density/distribution (Hartford et al., 2017; Shao et al., 2025), RKHS embeddings (Singh, Sahani, and Gretton, 2019; Bruns-Smith, 2025), and in our case, the conditional mean of the known transformation of the observations. Several papers iteratively solve the nested problem, despite the computational burden, for adversarial or minimax formulations (Lewis and Syrgkanis, 2018; Bennett, Kallus, and Schnabel, 2019; Dikkala et al., 2020). This computational advantage can be particularly valuable when employing machine learning methods that are often costly to train.

In the simulation study, we examine the finite sample performance of our Wald tests for point evaluations and average derivatives. We find that our method works well with different first-stage learners and in different scenarios. The tests maintain size close to the nominal level under the null. The power decreases as the number of instruments increases, particularly in smaller samples, but improves with larger sam-

ple sizes as expected. We also find that the performance of our method is particularly sensitive to noise the endogenous regressor that is not explained by the instruments, especially when the number of instruments is large. Some configurations also exhibit numerical instability, highlighting practical limitations of the method under weaker identification.

In the empirical application, we revisit the study of migration and economic mobility in Tanzania by Beegle, De Weerdt, and Dercon (2011). We apply nonparametric methods in both stages of one of their 2SLS specifications to estimate the relationship between migration distance and consumption growth. The application involves six excluded instruments, making conventional tensor-product sieve estimation computationally prohibitive. We find that the positive relationship the authors found is actually close to zero over much of the migration-distance distribution but increases sharply at longer distances.

This paper contributes to a growing literature incorporating machine learning methods into CMR models. Existing approaches built around particular learners include neural networks (Hartford et al., 2017; Bennett, Kallus, and Schnabel, 2019; Shao et al., 2025), forests (Athey, Tibshirani, and Wager, 2019; Oprescu, Syrgkanis, and Wu, 2019), and reproducing kernel Hilbert space (Singh, Sahani, and Gretton, 2019; Muandet et al., 2020), among many others. Much of this literature emphasizes flexible second-stage estimation of the structural function. In contrast, our focus is on accommodating flexible machine learning methods in the first-stage. Related literature studies machine learning for instrument selection (Belloni et al., 2012) and the construction of instruments through flexible first-stage prediction (Chen, Chen, and Lewis, 2020). More recently, Bruns-Smith (2025) places first-stage prediction at the center of NPIV estimation and relates structural estimation error to first-stage prediction error. Our work complements this line of research by exploiting the affine structure to formulate first-stage estimation as conditional mean prediction.

Our work also contributes to the literature on Neyman-orthogonalization and debiased learning (Chernozhukov et al., 2018, 2022; Foster and Syrgkanis, 2023). Drawing on Foster and Syrgkanis (2023) who studied the orthogonalization of general loss functions, we orthogonalize the sieve minimum distance criterion function. This

orthogonalization can be equivalently performed by setting the functional of interest as the criterion function in the framework of Chernozhukov et al. (2022). On top of the usual rate gain from the Neyman orthogonality, this construction allows us to derive an asymptotic linear representation whose form is agnostic to the choice of first-stage learner in the minimum distance framework.

A closely related paper is Bruns-Smith (2025), which also emphasizes flexible first-stage estimation in NPIV models. Bruns-Smith constructs a data-adaptive RKHS from a reduced-form prediction of the outcome and uses this learned representation for second-stage estimation. Our approach differs in several aspects. First, we consider the broader class of affine CMR models, where a natural reduced-form problem could be unclear. Second, we directly estimate the conditional means with any learner, while it is important in the framework of Bruns-Smith (2025) that the chosen machine learning method can produce a sufficiently rich RKHS. Finally, Bruns-Smith (2025) allows machine learning methods in both stages, whereas we retain a linear sieve in the second stage. This compromises the computational scalability of our second stage, but allows us to access the extensive literature on challenging inference problems built on sieve methods. Bruns-Smith (2025) instead provides inference for regular functionals through a separate debiasing step.

Our analysis points to several directions for future research. First, the sieve basis functions, which are the first-stage prediction outcomes, have a special structure. This motivates the development of machine learning methods and theory that can fully exploit the relationships among these outcomes. Second, the demanding first-stage rate requirements motivate the development of complementary tools to improve convergence rates, such as penalization and shape constraints. Finally, our simulation results suggest that the conditional mean implementation could be particularly sensitive to the noise in the endogenous regressor, perhaps due to the interaction between first-stage prediction errors and potentially increased ill-posedness. This motivates the development of more robust frameworks that can mitigate the amplification of first-stage prediction errors through ill-posed directions.

The remainder of the paper is organized as follows. Section 2 introduces affine CMR models with examples. Section 3 develops the proposed estimator and Wald

inference for smooth functionals. Section 4 examines first-stage machine learning methods and sufficient rate requirements for inference. Section 5 presents Monte Carlo simulations, and Section 6 provides an empirical application. Appendix A establishes a Gaussian coupling of the sieve coefficients. Appendices B and C contain the proofs and lower-level verification of the first-stage conditions, respectively.

**Notation.** For a measurable function $f$, let $\|f\|_{P,2} = (\mathbb{E}_P[f^2])^{1/2}$ and $\|f\|_{P,\infty} = \operatorname{ess\,sup} |f|$, where ess sup denotes the essential supremum under $P$. For vectors, $\|\cdot\|_2$ denotes the Euclidean norm, and for matrices, $\|\cdot\|_{\text{op}}$ denotes the operator norm. We write $a_n \precsim b_n$ if $a_n \leq C b_n$ for some constant $C > 0$ independent of $n$, and $a_n \asymp b_n$ if both $a_n \precsim b_n$ and $b_n \precsim a_n$. When uniformity over $P \in \mathcal{P}$ is asserted, the constants are also independent of $P$.

# 2 Affine CMR models

In this section, we formally introduce the affine CMR model and discuss its reduction in our problem with examples. The residual function in (1) is affine in the structural parameter $\theta$ if

$$\rho(O_i, \theta) \;\; = \;\; c(O_i) + \mathcal{L}[\theta; O_i]$$

for some known function $c(\cdot)$ and a known linear functional $\mathcal{L}[\cdot; o] : \mathcal{H} \to \mathbf{R}$. We assume the residual function $\rho(\cdot, \theta)$ is $P$-measurable for each $\theta \in \Theta \subset \mathcal{H}$ and $P \in \mathbf{P}$ where $\mathcal{H}$ is an Hilbert space equipped with a norm $\|\cdot\|_{\mathcal{H}}$ and $\Theta$ is a set of admissible parameter space.

We parametrize $\Theta$ with a linear sieve. That is, for some choice of sieve basis functions $\{p_{nl}(x)\}_{l=1,...,k_n} \subset \mathcal{H}$, we define a linear span $\mathcal{V}_n = \text{span}\{p_{n1}(x), ..., p_{nk_n}(x)\}$ and define the sieve space $\Theta_n \subset \Theta \cap \mathcal{V}_n$ as

$$\Theta_n \;\; := \;\; \{p_n(x)^\top \beta : \beta \in \mathcal{B}_n \subset \mathbf{R}^{k_n}\},$$

which grows dense in $\Theta$. Here, $\mathcal{B}_n$ is a set of feasible coefficients, which can reflect, for example, shape constraints or regularizations. For convenience, after replacing all nonparametric parts with its sieve counterparts, we rewrite the residual function as a linear function of $c_n$ and $q_n := \mathcal{L}[p_n(\cdot); O_i]$ thanks to the linearity of $\mathcal{L}$ as follow:

$$\rho(O_i, \theta_n) = c(O_i) + q_n(O_i)^\top \beta_n. \tag{3}$$

Taking the conditional expectation with respect to instruments $Z$ yields the moment equation in Equation (2).

*Remark* 1. It is straightforward to extend the above formulation to a fixed-dimensional Euclidean parameter $b \in \mathbf{R}^{d_b}$ and a fixed number of nonparametric components $h_j \in \mathcal{H}_j$, $j = 1, ..., J$. Let $\theta = (b, h_1, \ldots, h_J) \in \mathbb{R}^{d_b} \times \mathcal{H}_1 \times \cdots \times \mathcal{H}_J$. The residual function takes the form

$$\rho(O, \theta) = c(O) + a(O)^\top b + \sum_{j=1}^{J} L_j[h_j; O]$$

where $L_j[\cdot; O]$, $j = 1, ..., J$ are known linear functionals. Approximating each $h_j$ by $p_{j,n}^\top \gamma_{j,n}$ and stacking the sieve coefficients and the corresponding basis functions again yields the expression in Equation (3) with $k_n = d_b + \sum_{j=1}^{J} k_{j,n}$ unknown Euclidean parameters. See Chen and Pouzo (2015) for the precise product-space construction of semi/nonparametric conditional moment models.

**Example 2.1** (Nonparametric instrumental variable models; NPIV)**.** NPIV is a popular model in empirical research. The recent work of Compiani (2022) and Denteh and Liebert (2026) respectively estimate the demand of strawberry markets and the conditional average treatment effect of a randomized policy by using NPIV and partially linear models. NPIV model specifies an unknown structural function $h_0$ through the moment equation $\rho(Y, X, h_0) = Y - h_0(X)$ where $Y$ is an outcome, $X$ a vector of regressors, and $Z$ a vector of instruments. After replacing the structural function $h_0$ with a sieve approximate $h_n(x) = p_n(x)^\top \beta_n$, the residual function fits the formulation in (3) with $c_n(O) = Y$ and $q_n(O) = -p_n(X)$.

**Example 2.2** (Transformation IV)**.** Florens and Sokullu (2017) consider nonparametric models on both sides of the equations.

$$H_0(Y) = \varphi_0(D) + X^\top b_0 + U, \qquad \mathbb{E}[U \mid X, W] = 0,$$

where $H_0$ and $\varphi_0$ are unknown functions, $D$ is an endogenous regressor, $X$ is a vector of exogenous regressors, and $W$ is a vector of instruments. Set $Z = (X, W)$. For $\theta = (b, H, \varphi)$, $\rho(O, \theta) = H(Y) - \varphi(D) - X^\top b$, $c(O) = 0$, $a(O) = -X$. With sieve approximates $H_n(y) = p_H(y)^\top \beta_H$ and $\varphi_n(d) = p_\varphi(d)^\top \beta_\varphi$, $q_n(O) = (-X^\top, p_H(Y)^\top, -p_\varphi(D)^\top)^\top$, $\beta_n = (b^\top, \beta_H^\top, \beta_\varphi^\top)^\top$, we have $\rho(O, \theta_n) = q_n(O)^\top \beta_n$.

**Example 2.3** (Linear CMR)**.** Bennett et al. (2023) consider the conditional moment restriction that is linear in the structural function of interest $h$

$$\mathbb{E}[g_1(O)h_0(X) \mid Z] = \mathbb{E}[g_2(O) \mid Z],$$

where $g_1$ and $g_2$ are known functions of the observed data and $h_0$ is unknown. Hence, we may set $c(O) = -g_2(O)$ and $q_n(O) = g_1(O)p_n(X)$ after substituting $h_n(x) = p_n(x)^\top \beta_n$ in the residual equation.

**Example 2.4** (Euler equation)**.** Escanciano et al. (2021) consider the consumption based asset pricing Euler equation

$$\mathbb{E}[b_0 R_{t+1} g_0(Z_{t+1}) \mid Z_t] - g_0(Z_t) = 0, \qquad Z_t = (C_t, V_t),$$

where $C_t$ is consumption, $V_t$ contains other utility-relevant variables, $R_{t+1}$ is a gross asset return, $b_0$ is the discount factor, and $g_0$ is marginal utility. For a fixed discount factor, the residual is linear in $g$. When both $b$ and $g$ are unknown, we may introduce $f = bg$ and write $\theta = (f, g)$. Then the residual becomes $\rho(O_t, \theta) = R_{t+1} f(Z_{t+1}) - g(Z_t)$, which is again linear in $(f, g)$. The original model is retained by imposing $f = bg$ for some $b \in (0, 1)$ as a restriction. With the sieve approximates $f_n(z) = p_n(z)^\top \gamma$ and $g_n(z) = p_n(z)^\top \beta$, we can set $q_n(O_t) = (R_{t+1} p_n(Z_{t+1})^\top, -p_n(Z_t)^\top)^\top$ and $\beta_n = (\gamma^\top, \beta^\top)^\top$. We leave this example for future research, as it requires time-

series generalization. See, for example, Chen, Liao, and Wang (2025) for time-series generalization of (nonlinear) sieve minimum distance estimators.

# 3 Estimation and Inference

This section develops estimation and inference for affine conditional moment restriction models. We focus on the affine model with a single nonparametric function for better exposition. We first introduce an orthogonalized sieve minimum distance estimator and establish an asymptotic linear representation of its coefficients under high-level first-stage prediction-error conditions. Building on this representation, we develop Wald inference for smooth functionals of the structural parameter.

## 3.1 Orthogonalized Sieve Minimum Distance Estimator

The conventional (population) plug-in minimum distance criterion function is

$$Q_P^{plug-in}(\theta) \quad = \quad \mathbb{E}_P\Big[\frac{m(Z,\theta)^2}{\sigma^2(Z)}\Big],$$

where $\sigma^2(z)$ is a weighting function. Since the true structural function $\theta_P$ satisfies the moment condition $m(Z,\theta_P)=0$, it is a natural minimizer of $Q_P(\theta)$. By the law of iterated expectations, the criterion function can be equivalently expressed as

$$Q_P(\theta) \quad = \quad \mathbb{E}_P[\frac{2\rho(Z,\theta)m(Z,\theta)-m(Z,\theta)^2}{\sigma^2(Z)}].$$

We call the second criterion function as an orthogonalized criterion function. Through out the analysis, we treat the weighting function $\sigma^2(\cdot)$ as known[1] and leave the optimal weighting scheme to future work. We denote the sieve minimizer of the ortogonalized criterion function as $\theta_{P,n} := \text{argmin}_{\Theta_n} Q_P$ and the associated sieve coefficients as $\beta_{P,n}$, so that $\theta_{P,n}(x) = p_n(x)^\top \beta_{P,n}$.

[1]A typical choice of $\sigma^2(z)$ is $\text{Var}_P(\rho(O,\theta_P)|z)$. It can be estimated by $\text{Var}_n(\rho(O_i,\tilde{\theta}_n)|Z_i)$ with a preliminary estimator $\tilde{\theta}_n$, for instance, without weighting.

The above criterion function is orthogonal in the sense of Foster and Syrgkanis (2023). That is, the gradient of the criterion function is Neyman-orthogonal to perturbations in the estimated nuisance, which is the conditional moment in our case. As a result, the first stage prediction error only appears in second-order and the estimators enjoy the same rate gain as in Chernozhukov et al. (2018). The orthogonalized criterion function can also be constructed through a first-step influence function correction by taking the functional of interest as the plug-in criterion function in the framework of Chernozhukov et al. (2022), or as an antiderivative of an orthogonal moment condition of Chernozhukov et al. (2018).

The sieve minimum distance estimator is defined as the minimizer of $\hat{Q}_n$, the sample counterpart of the criterion function, over the sieve space $\Theta_n$:

$$\hat{\theta}_n \quad := \quad \underset{\Theta_n}{\operatorname{argmin}}\, \hat{Q}_n(\theta) = \underset{\Theta_n}{\operatorname{argmin}} \left\{ \frac{1}{n} \sum_{i=1}^{n} \frac{2\hat{m}_n(Z_i,\theta)\rho(Z_i,\theta) - \hat{m}_n(Z_i,\theta)^2}{\sigma^2(Z_i)} \right\}$$

Likewise, we denote the sieve coefficients as $\hat{\beta}_n$, so that $\hat{\theta}_n(x) = p_n(x)^\top \hat{\beta}_n$. In the rest of the paper, when evaluating the residual, conditional moment, or criterion function at a parameter $\theta_n$ on the sieve space, we sometimes use the sieve coefficients directly as arguments, for instance, $Q_P(\beta_n) = Q_P(p_n^\top \beta_n)$.

*Remark* 2 (Orthogonal projection first-stage). Suppose the conditional moment is estimated by orthogonal projection methods such as sieve projections (Chen and Pouzo, 2015). Let $R$ denote the vector of residuals $\rho(O_i, \cdot\ )$ and $\Pi$ denote the orthogonal projection matrix. Ignoring weighting for simplicity,

$$\hat{Q}_n(\theta) = \frac{1}{n}\{2R^T(\Pi R) - R^\top \Pi^\top \Pi R\} = \frac{1}{n} R^\top \Pi^\top \Pi R = \frac{1}{n} \sum_i \hat{m}(Z_i,\theta)^2,$$

due to the idempotency of $\Pi$. Thus, the orthogonalized criterion reduces to the conventional plug-in criterion function. However, such reduction may not generally extend to other first-stage learners, such as kernel, nearest neighbors, or tree/forest estimators.

We now introduce the assumptions underlying our asymptotic analysis and discuss each component of the estimator in detail. We begin with regularity conditions on the data-generating process and the sieve approximation, followed by conditions on first-stage estimation and the population criterion.

**Assumption 3.1** (DGP)**.** (i) $\{O_i\}_{i=1}^n$ are i.i.d., $O_i \sim P \in \mathbf{P}$. (ii) There exists a constant $C_\rho, C_c < \infty$ such that $\sup_P \sup_\theta \|\mathbb{E}_P[\rho(O,\theta)^2|Z]\|_{P,\infty} \leq C_\rho$ and $\sup_{P\in\mathcal{P}} \|E_P[c(O)^2|Z]\|_{P,\infty} \leq C_c$. (iii) For every $P \in \mathbf{P}$, there exists $\theta_P \in \Theta$ such that $\mathbb{E}[\rho(O;\theta_P)|Z] = 0$ for $P$-almost every $Z$. (iv) There exists a constant $C_\mathcal{L} < \infty$, such that for every $P \in \mathbf{P}$, $\|\mathcal{L}[h;\cdot]\|_{P,2} := \{E_P[\mathcal{L}[h;\cdot]^2]\}^{1/2} \leq C_\mathcal{L}\|h\|_\mathcal{H}$ for all $h \in \mathcal{H}$ and $\|\mathcal{L}[h;\cdot]\|_{P,\infty} := ess\sup_o |\mathcal{L}[h;\cdot]| \leq C_\mathcal{L}\|h\|_\infty$ for all $h \in \mathcal{H}$ with $\|h\|_\infty < \infty$. (v) For some constant $M < \infty$, $\sup_{\theta\in\Theta} \|\theta\|_\mathcal{H} \leq M$. (vi) There exist constants $0 < c \leq C \leq \infty$ such that for every $P \in \mathbf{P}$, $c\|h\|^2_{L_2(P_X)} \leq \|h\|^2_\mathcal{H} \leq C\|h\|^2_{L_2(P_X)}$ for all $h \in \mathcal{H}$.

**Assumption 3.2** (Sieve)**.** (i) Let the Gram matrix be $G_n := \mathbb{E}_P[p_n(X)p_n(X)^\top]$. There exist constants $\underline{\lambda}$ and $\overline{\lambda}$ such that for every $n$ and $P \in \mathbf{P}$, $0 < \underline{\lambda} \leq \lambda_{\min}(G_n) \leq \lambda_{\max}(G_n) < \overline{\lambda} < \infty$. (ii) There exists a sequence $\xi_n$ such that for every $n$ and $P \in \mathbf{P}$, $\operatorname{ess\,sup}_x \|p_n(x)\|_{l_2} \leq \xi_n$. (iii) For every $\theta \in \Theta$, $\inf_{\theta_n\in\Theta_n} \|\theta - \theta_n\|_\mathcal{H} = O(r_{s,n}) = o(1)$.

Assumption 3.1(iv) translates the primitive Assumption 3.2(ii) on the sieve basis functions to $q = L_O(p)$, which is actually used in analysis. In the NPIV example, Assumption 3.1(iv) trivially holds with constant 1 for the sup-norm and $c^{-1/2}$ for the $L_2$ norm due to Assumption 3.1(vi). Assumption 3.1(v) is mild and standard in the literature. In particular, we assume the constant $M$ is large enough, so that the minimizers of the criterions are always interior solutions. A convenient sufficient condition for Assumption 3.1(vi) is that the density of $X$ is bounded both from below and above.

For theoretical convenience, we orthonormalize the sieve basis with respect to $\|\cdot\|_\mathcal{H}$ without loss of generality. Together with Assumption 3.1(vi), we may obtain $C^{-1}I_{k_n} \preceq \tilde{G}_n \preceq c^{-1}I_{k_n}$ where $\tilde{G}_n$ is the $\|\cdot\|_\mathcal{H}$-orthonormalized Gram matrix. In Assumption 3.2(ii), $\xi_n$ is $O(\sqrt{k_n})$ for many of the popular choices of sieves, including B-splines. Assumption 3.2(iii) is a standard sieve approximation assumption and the

rate is readily available for popular choices of sieve (e.g., Chen, 2007).

By the affine representation in Equation (2), we can construct the conditional moment at any sieve coefficient $\beta_n$ from the conditional means $\mu_c(z) = \mathbb{E}_P[c(O)|Z = z]$ and $\mu_q(z) = \mathbb{E}_P[q_n(O)|Z = z]$. We estimate these conditional means using flexible machine learning methods. We employ $L$-fold cross-fitting to avoid a potential bias from self-prediction. That is, we partition the sample into $L$ disjoint folds and estimate the conditional means for each fold using observations from the remaining folds (Chernozhukov et al., 2018). For each observation $i$ in fold $\ell$, we evaluate the conditional moment as

$$\hat{m}_n(Z_i, \beta) = \hat{\mu}_c^{(-\ell)}(Z_i) + \hat{\mu}_q^{(-\ell)}(Z_i)^\top \beta,$$

with the conditional means $\hat{\mu}^{(-\ell)}(\cdot)$ are learned on the folds that excludes $\ell$. The conditional mean estimators are always obtained by cross-fitting, so we often omit the superscript $(-\ell)$ when the context is clear.

**Assumption 3.3** (first-stage)**.** For an independent draw of $Z \sim P_Z$, there exists deterministic sequences $r_{\mu,n}$, $r_{m,n} \to 0$ such that, $\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2|\mathcal{T}_\ell] = O_P(r_{\mu,n}^2)$ and $\mathbb{E}_P[\{\hat{m}_n^{(-\ell)}(Z, \beta_{P,n}) - m_{n,P}(Z, \beta_{P,n})\}^2|\mathcal{T}_\ell] = O_P(r_{m,n}^2)$.

Although we obtain the conditional moment $\hat{m}_n(z, \cdot)$ through the estimated conditional means $\hat{\mu}_n(z)$, the regularization on $m(z, \cdot)$ can significantly reduce the rate $r_{m,n}$ relative to $r_{\mu,n}$. Hence, we state the rates separately for future generalization. We discuss the verification of the above assumption using finite sample error bounds in the literature in Section 4.

*Remark* 3. We mainly concern machine learning methods that can predict vector outcomes. However, $k_n$-many separate scalar learners can still be used. Suppose individual learners satisfy $\mathbb{E}_P[\int |\hat{\mu}_{n,j}^{(-\ell)}(z) - \mu_{n,P,j}(z)|^2 dP_Z|\mathcal{T}_\ell] \leq b_{n,j}^2$ for $j = 1, ...k_n$. Summing up the componentwise errors, the vector mean squared error is bounded by $\sum_{j=0}^{k_n} b_{n,j}^2$. If there is an $a_n$ such that $b_{n,j}^2 \leq C a_n^2$ for each $j = 0, ..., k_n$ accounting for $k_n \to \infty$, then $r_{\mu,n}$ can be taken as $\sqrt{k_n + 1} a_n$.

In population, the orthogonalized criterion function coincides with the conventional minimum distance criterion by the law of iterated expectations. Hence, the orthogonalization does not alter identification. The identification in affine model is particularly transparent. For $h \in \mathcal{H}$, define the weak norm as

$$\|h\|_w^2 := \mathbb{E}_P \left[ \frac{\{\mathbb{E}_P[\mathcal{L}[h; O] \mid Z]\}^2}{\sigma^2(Z)} \right].$$

Thanks to the affinity of the moment and the moment condition $m(\cdot, \theta_P) = 0$, it holds that $m(\cdot, \theta) = m(\cdot, \theta - \theta_P)$. Hence, the population criterion satisfies

$$Q_P(\theta) = \mathbb{E}_P \left[ \frac{m(Z, \theta)^2}{\sigma^2(Z)} \right] = \|\theta - \theta_P\|_w^2.$$

Assumption 3.4(i) below imposes point identification of $\theta_P$. In the affine model, this condition is equivalent to injectivity of the conditional mean operator $h \mapsto \mathbb{E}_P[\mathcal{L}[h; O] \mid Z]$ on the set of admissible deviations $\{\theta - \theta_P : \theta \in \Theta\}$. In the NPIV model, this reduces to the usual completeness condition (Newey and Powell, 2003).

**Assumption 3.4** (Criterion function)**.** (i) For every $P \in \mathbf{P}$ and $\theta \in \Theta$, $Q_P(\theta) = 0$ if and only if $\|\theta - \theta_P\|_{\mathcal{H}} = 0$. (ii) There is a sequence $r_{s,n}$ and $\Pi_n \theta_P \in \Theta_n$ such that $\|\theta_P - \Pi_n \theta_P\|_{\mathcal{H}} = O(r_{s,n}) = o(1)$. (iii) For each $P \in \mathbf{P}$, there is a sequence $r_{w,P,n}$ such that $Q_P(\Pi_n \theta_P) = \|\Pi_n \theta_P - \theta_P\|_w^2 \leq O(r_{w,P,n}^2) = o(1)$. (iv) There exist constants $0 < \underline{\sigma}^2 \leq \bar{\sigma}^2 < \infty$, such that for every $P \in \mathbf{P}$, $\underline{\sigma}^2 \leq \sigma^2(Z) \leq \bar{\sigma}^2$ $P$-a.s..

Assumption 3.4(ii) follows from the sieve approximation condition in Assumption 3.2(iii). Assumption 3.4(ii) also implies Assumption 3.4(iii) with $r_{w,P,n} = r_{s,n}$ by Jensen's inequality, Assumptions 3.1(iv), and 3.4(iv). We state the weak approximation rate separately to allow for sharper bounds. Notice Assumption 3.4(ii)(iii) are on the sieve approximation of the true structural function $\Pi_n \theta_0$, which in general differ from the sieve minimizer $\theta_{P,n}$.

Following Chen and Pouzo (2015), we define the sieve measure of ill-posedness.

For a sieve direction $v = p_n^\top b \in \mathcal{V}_n$, let

$$H_n := \mathbb{E}_P\left[\frac{\mu_q(Z)\mu_q(Z)^\top}{\sigma^2(Z)}\right], \qquad \tau_n^2 := \sup_{b\neq 0}\frac{b^\top G_n b}{b^\top H_n b},$$

where $G_n$ is defined in Assumption 3.2(i). On the sieve space, the weak norm satisfies $\|v\|_w^2 = b^\top H_n b$ and $\|v\|_{L_2(P_X)} \leq \tau_n \|v\|_w$. If the conditional moment operator is injective on $\mathcal{V}_n$, then $H_n$ is positive definite for each $n$, although its smallest eigenvalue may converge to zero as $n \to \infty$. Because the conditional moment is affine in $\theta$, its directional derivative and the matrix $H_n$ do not depend on the evaluation point. Hence, we do not distinguish the local and global sieve measures of ill-posedness. See Chen and Pouzo (2015) for the local measure in nonlinear models, and Newey and Powell (2003), Carrasco, Florens, and Renault (2007), Chen and Christensen (2018), and Chen, Christensen, and Kankanala (2025) for treatments of ill-posedness in NPIV and CMR models.

Now we derive the asymptotic linear representation of the sieve coefficients, which is useful for subsequent inference. Differentiating the population criterion function, we obtain the gradient and Hessian matrix

$$\begin{aligned}
\Psi_{n,P}(\theta_n) &:= \frac{1}{2}\nabla_\beta Q_P(\theta_n) \\
&= \mathbb{E}_P\left[\frac{\rho(O,\theta_n)\mu_q(Z) + m(Z,\theta_n)\{q_n(O) - \mu_q(Z)\}}{\sigma^2(Z)}\right], \qquad (4)\\
\nabla_\beta \Psi_{n,P}(\theta_n) &= \mathbb{E}_P\left[\frac{\mu_q(Z)\mu_q(Z)^\top}{\sigma^2(Z)}\right].
\end{aligned}$$

Notice the population Hessian matrix coincide with $H_n$ defined above in the weak norm. Let $\psi_n(O;\theta)$ denote the integrand in the expression for $\Psi_{n,P}$. Similarly, differentiating the cross-fitted sample criterion function yields

$$\hat{\Psi}_n(\beta) := \frac{1}{n}\sum_{i=1}^n \hat{\psi}_{ni}(\beta), \quad \text{where} \quad \hat{\psi}_{ni}(\beta) = \frac{\rho(O_i,\beta)\hat{\mu}_q(Z_i) + \hat{m}_n(Z_i,\beta)(q_n(O_i) - \hat{\mu}_q(Z_i))}{\sigma(Z_i)^2}.$$

The corresponding sample Hessian matrix is

$$\hat{H}_n := \nabla_\beta \hat{\Psi}_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} \frac{q_n(O_i)\hat{\mu}_q(Z_i)^\top + \hat{\mu}_q(Z_i)(q_n(O_i) - \hat{\mu}_q(Z_i))^\top}{\sigma(Z_i)^2}.$$

*Remark* 4 (Orthogonal projection first-stage continued). If the first-stage estimator is an orthogonal projection estimator, ignoring the weighting and sample-splitting, the sample average of the second part of $\hat{\Psi}_n$ vanishes because $\hat{m}$ belongs to the spanned IV space and $(q - \hat{\mu})$ belongs to the null space of the spanned IV space. Hence, the gradient of the criterion function again reduces to its counterpart in Chen and Pouzo (2015). This reduction extends to smooth nonlinear models beyond affine settings, but may not extend to nonprojection learners.

*Remark* 5 (Comparison to DML). As expected from Foster and Syrgkanis (2023), the first order condition of the orthogonalized criterion function can work as the orthogonal moment condition for GMM (Chernozhukov et al., 2018, 2022). Hence, the orthogonalized minimum distance estimator here and the GMM estimator of double machine learning can coincide with a particular choice of weighting functions. However, they in general differ when the parameter space is restricted. The comparison between minimum distance and GMM estimator goes back to, at least, Newey and McFadden (1994).

Our first order expansion relies on the inverse of the estimated Hessian $\hat{H}_n$. Lemma B.2 shows that $\hat{H}_n$ is indeed positive definite (and hence invertible) with probability approaching one under the rate requirement in Lemma 3.1. Provided the inverse of $\hat{H}_n$ and the interior solution $\hat{\beta}_n$, the following holds without remainder, due to the affinity of the moment.

$$\hat{\beta}_n - \beta_{P,n} = \hat{H}_n^{-1}\{\hat{\Psi}_n(\hat{\beta}_n) - \hat{\Psi}_n(\beta_{P,n})\} = -\hat{H}_n^{-1}\hat{\Psi}_n(\beta_{P,n}). \tag{5}$$

The last equality uses $\hat{\Psi}_n(\hat{\beta}_n) = 0$ and $\hat{H}_n$ is positive definite. Hence, we derive a natural linear representation of the sieve coefficients with the population version of Equation (5) as a leading term.

**Lemma 3.1.** Suppose Assumptions 3.1, 3.2, 3.3, and 3.4(iii)(iv) hold. Moreover, suppose

$$\tau_n^2 r_{H,n} := \tau_n^2 \Big( \xi_n \sqrt{\frac{\log k_n}{n}} + \frac{r_{\mu,n}\xi_n}{\sqrt{n}} + r_{\mu,n}^2 \Big) = o(1). \tag{6}$$

Then,

$$\sqrt{n}(\hat{\beta}_n - \beta_{P,n}) = -H_{n,P}^{-1} \frac{1}{\sqrt{n}} \sum_i \psi_n(O_i, \theta_{P,n}) + R_n$$

where $\|R_n\|_2 = O_p(a_n + b_n)$ with

$$\begin{aligned} a_n &= \tau_n^2 \{ \xi_n \sqrt{\frac{\log k_n}{n}} + \frac{r_{\mu,n}\xi_n}{\sqrt{n}} + r_{\mu,n}^2 \} \{ \tau_n \sqrt{k_n} + \tau_n^2 \xi_n r_{w,n} \}, & (7) \\ b_n &= \tau_n^2 \{ r_{\mu,n} + \xi_n r_{m,n} + \sqrt{n} r_{m,n} r_{\mu,n} \}. & (8) \end{aligned}$$

The rate requirement $\tau_n^2 r_{H,n} = o(1)$ is a sufficient condition for the Hessian inversion. It is mild and dominated by the rate requirements implied by $a_n$ and $b_n$. The rates in $a_n$ and $b_n$ are the convergence rates of the remainders from replacing the estimated term in Equation (5) to population terms. Hence, they can be captured by the prediction error rate $r_{\mu,n}$ in Assumption 3.3.

*Remark* 6 (Comparison to CP score). Evaluating $\psi_n$ at the true structural parameter gives $\psi_n(O; \theta_P) = \rho(O, \theta_P)\mu_q(Z)/\sigma^2(Z)$, which corresponds to the leading stochastic term in Chen and Pouzo (2015, Equation (3.11)). The distinction lies in how this leading term is obtained from the sample criterion. The gradient of the plug-in minimum distance criterion is $n^{-1}\sum_{i=1}^n \hat{m}_i(\theta)\hat{\mu}_{q,i}/\sigma^2(Z_i)$, so replacing $\hat{m}_i$ with $\rho_i$ requires a separate justification. The orthogonalized criterion places $\rho_i$ directly in the gradient, and its difference from the population score can be controlled through the prediction errors for any first-stage learners.

As a final remark, the analysis on the orthogonalized criterion function can be extended to smooth nonlinear CMR models beyond affine settings, provided appropriate estimators of the conditional moment and its derivative are available (e.g.,

Chen and Pouzo, 2015). Estimating the conditional distribution in the first stage seems particularly attractive for constructing these objects in general settings (Hartford et al., 2017; Shao et al., 2025). We leave such generalization to future work. Building on the linear representation in Lemma 3.1, we provide a Wald statistic for testing smooth functionals in the following subsection. In Appendix A, we provide a Gaussian coupling of the linear representation, which is useful for various inference problems.

## 3.2 Inference for smooth functionals

We provide a Wald statistic for smooth functionals $\phi$ that covers a lot of interesting inference problems in empirical research. Particularly, we modify the Wald statistic of Chen and Pouzo (2015) to our setting. Consider the null hypothesis

$$H_0 : \phi[\theta_0] = \phi_0.$$

For smooth functionals $\phi[\cdot]$, their first order behavior on the sieve space admits a convenient expression in terms of coefficient loadings $l_n$ (Newey, 1997). The coefficient loadings can be defined as $l_n := (\dot{\phi}[p_{n1}], \cdots, \dot{\phi}[p_{nk_n}])^\top$ for some linear functional $\dot{\phi}[\cdot] : \mathcal{H} \to \mathbf{R}$. If $\phi$ is directionally differentiable, $\dot{\phi}[\cdot]$ can be taken as the directional derivative of $\phi$ at $\theta_P$. Note the Riesz representer $v_n^*$ of the functioanl $\phi$ in Chen and Pouzo (2015) corresponds to $p_n(x)^\top H_n^{-1} l_n$.

**Example 2.1** (NPIV, continued)**.** Suppose an empirical researcher estimates a demand function using the NPIV model $Y = \theta_0(X) + U$, where $Y$ is (log) quantity and $X$ contains (log) prices. A researcher may be interested in demand at price $x_0$ and its confidence interval. On the sieve space, the testing functional is written as

$$\phi(\theta_{P,n}) = \theta_{P,n}(x_0) = p_n(x_0)^\top \beta_{P,n}.$$

Hence, the loading $l_n$ is taken as $p_n(x_0)$.

**Example 2.1** (NPIV, continued)**.** Continuing from the previous example, a re-

searcher may be interested in the (weighted) average price elasticity of its own price. Let $x_1$ denote the log of the own price. Then, for some known weights $w(x)$,

$$\phi(\theta_{P,n}) = \int \frac{\partial \theta_{P,n}(x)}{\partial x_1} w(x)\, dx = \left\{ \int \frac{\partial p_n(x)}{\partial x_1} w(x)\, dx \right\}^\top \beta_{P,n},$$

Hence, the loading $l_n$ can be taken as $l_n = \int \frac{\partial p_n(x)}{\partial x_1} w(x)\, dx$.

**Example 2.1** (NPIV, continued)**.** Continuing from the previous example, a researcher may instead be interested in demand in levels at $x_0$. Then, the testing functional is

$$\phi(\theta_{P,n}) = \exp\{\theta_{P,n}(x_0)\} = \exp\{p_n(x_0)^\top \beta_{P,n}\}.$$

Since $\phi(\cdot)$ is smooth, it can be approximated by the Tayler expansion near $\beta_{P,n}$:

$$\phi(\theta_n) = \exp\{p_n(x_0)^\top \beta_{P,n}\} + \nabla_\beta \exp\{p_n(x_0)^\top \beta_{P,n}\}(\beta_n - \beta_{P,n}) + R,$$

where the remainder $R$ is of the second order in $\|\beta - \beta_{P,n}\|_2$ (e.g., Magnus and Neudecker, 2019, Chapter 6, Theorem 12). Since $\beta_{P,n}$ is unknown, one may estimate the loading as $l_n = \nabla_\beta \exp(p_n(x_0)^\top \hat{\beta}_n))$.

Provided the coefficient loading $l_n$ and the linear expansion in Lemma 3.1, we can write (the numerator of) the test statistic as

$$\sqrt{n}(\phi(\hat{\theta}_n) - \phi(\theta_P)) \quad \approx \quad -l_n^\top H_n^{-1} \frac{1}{\sqrt{n}} \sum_i \psi_n(O_i) + l_n^\top R_{n,P}. \tag{9}$$

This suggests the sieve Wald score and its estimator

$$\begin{aligned} S_{ni} &= l_n^\top H_n^{-1} \frac{\rho_i(\beta_{P,n})\mu_q(Z_i) + m(Z_i, \beta_{P,n})(q(X_i) - \mu_q(Z_i))}{\sigma_i^2}, \\ \hat{S}_{ni} &= l_n^\top \hat{H}_n^{-1} \frac{\rho_i(\hat{\beta}_n)\hat{\mu}_q(Z_i) + \hat{m}(Z_i, \hat{\beta}_n)(q(X_i) - \hat{\mu}_q(Z_i))}{\sigma_i^2}. \end{aligned}$$

We define the standard error of the Wald statistic and its estimator

$$V_{n,P} := \mathrm{Var}_P[S_{ni}], \quad \hat{V}_{n,P} := \mathrm{Var}_n[\hat{S}_{ni}],$$

where $\mathrm{Var}_n(X) = \mathbb{E}_n[X^2] - (\mathbb{E}_n X)^2$. Equivalently, they can be written as $V_{n,P} = l_n^\top H_n^{-1} \Omega_n H_n^{-1} l_n$ where $\Omega_n := E_P[\psi_{n,P}(O)\psi_{n,P}(O)^\top]$, and $\hat{V}_n = l_n^\top \hat{H}_n^{-1} \hat{\Omega}_n \hat{H}_n^{-1} l_n$ where $\hat{\Omega}_n = \mathbb{E}_n[\hat{\psi}_i \hat{\psi}_i^\top]$. The consistency of $\hat{V}_n$ is provided in Lemma B.3. As the dimension of the sieve grows, $\|l_n\|_2$ may diverge depending on the testing functional. For details on such *irregular* functionals, see Chen and Liao (2014) and Chen and Pouzo (2015).

Following Chen and Pouzo (2015), we impose the following assumptions.

**Assumption 3.5** (Testing functional $\phi$)**.** (i) There exists $\mathcal{N}_{n,P} \subset \Theta_n$ and a linear functional $\dot{\phi} : \mathcal{V}_n \to \mathbf{R}$ such that $P\{\hat{\theta}_n \in \mathcal{N}_{n,P}\} \to 1$ and $\sup_{\theta \in \mathcal{N}_{n,P}} \frac{\sqrt{n}}{\sqrt{V_n}} |\phi(\theta) - \phi(\theta_{P,n}) - \dot{\phi}[\theta - \theta_{P,n}]| = o(1)$. (ii) $\frac{\sqrt{n}}{\sqrt{V_n}} |\phi(\theta_{P,n}) - \phi(\theta_P)| = o(1)$.

**Assumption 3.6** (Sieve score)**.** (i) (centering) $E_P[S_{ni}] = 0$. (ii) (Lindeberg or Lyapunov condition) $E_P[\frac{S_{ni}^2}{V_n} 1\{|S_{ni} > \epsilon \sqrt{n V_n}\}] \to 0$.

If $\phi$ is linear, then the absolute value in Assumption 3.5(i) is exactly 0 and $\mathcal{N}_{n,P}$ can be taken as entire $\mathbf{R}^{k_n}$. Otherwise, one may construct a local neighborhood near $\beta_{P,n}$ after establishing consistency and control the estimation and linearization errors across the local neighborhood. Assumption 3.5(ii) is a combined requirement on the sieve approximation bias and the testing functional. It sets the lower bound on the sieve complexity $k_n$. Assumption 3.6(i) is automatically satisfied because $\beta_{P,n}$ is always the interior solution, but we leave it for future generalization. A convenient sufficient condition for Assumption 3.2(ii) is the famous Lyapunov condition $E_P[|S_{ni}|^{2+\delta}/n^{\delta/2} V_n^{1+\delta/2}] \to 0$, we in turn can be satisfied by the $2+\delta$ moments of the residual, sieve basis, and conditional means.

**Theorem 3.1** (Wald)**.** Suppose Assumptions 3.1, 3.2, 3.3, 3.4(iii)(iv), 3.5, and 3.6 hold. Moreover, suppose the rate conditions $\tau_n^2 r_{H,n} = o(1)$, $\|l_n\|_2 (a_n + b_n)/\sqrt{V_n} = o(1)$

hold.

$$\frac{\sqrt{n}}{\sqrt{\hat{V}_n}}\{\phi(\hat{\theta}_n) - \phi(\theta_P)\} \rightsquigarrow N(0,1).$$

The first rate requirement $\tau_n^2 r_{H,n} = o(1)$ for Hessian convergence is mild. The second rate requirement is to control the remainder term in Equation (9), and is stronger than necessary. A convenient sufficient condition for the second rate requirement is $a_n + b_n = o(1)$, although the rate requirement can be relaxed up to $\tau_n^2$. We defer the discussion of the rate requirement to the following section.

# 4 First-stage learners and the rate requirements

This section surveys several machine learning methods and their prediction error bounds, followed by a discussion on the rate requirements of statistical inferences.

## 4.1 First stage learners and their error bounds

Recall Assumption 3.3 requires bounds on the prediction errors of the estimated conditional means and the resulting conditional moment. However, such bounds are often not directly available in the theory of machine learning methods. In particular, the increasing dimensionality of the prediction outcome $q_n(\cdot)$ is typically not a concern in the theory of first-stage learners. Therefore, we verify Assumption 3.3 for selected machine learning methods by adapting their derivation of the error bounds. The proofs are provided in Appendix C.

The first class of learners consists of local-smoothing learners such as kernel estimators. These methods exploit the smoothness of conditional means on shrinking neighborhoods of the conditioning point, while the neighborhoods are constructed differently across learners. Their convergence theory often provides pointwise prediction-error bounds. We impose additional regularity conditions when necessary and integrate these bounds over $P_Z$ to verify Assumption 3.3.

**Example 4.1** (Generalized random forest)**.** Athey, Tibshirani, and Wager (2019)

propose generalized random forest, which uses the grown trees to obtain kernel-like weights $\alpha_i$. For an evaluation point $z$ and in each tree, observations whose $Z_i$ falls in the same leaf as $z$ recieves a unit weight. Aggregating those weights over trees yields the final weight $\alpha_i(z)$. Our first-stage conditional means then can be estimated as the weighted average

$$\hat{\mu}_n(z) = \sum_i \alpha_i(z)(c_n(O_i), q_n(O_i)^\top)^\top.$$

Their Lemma 7 provides pointwise bias and variance bounds of $\hat{\mu}(z)$, which does not depend on $z$. Under some regularity conditions, such as the bounded second moment and the Lipschitz continuity of the conditional means, we may bound the first-stage error by aggregating the pointwise bound and obtain

$$\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2 \mid \mathcal{T}_\ell] = O_P\Big(L_{\mu,n}^2 s_n^{-2\gamma} + (1+\xi_n^2)\frac{s_n}{n}\Big).$$

Hence, we may take $r_{\mu,n}$ as the square root of the sequence in $O_p(\cdot)$ term. Here, $s_n$ is the subsampling size, which is a tuning parameter, $L_{\mu,n}$ is the Lipschitz constant of $\mu_{n,P}$, which may grow with $n$, $\xi_n$ is the sieve envelope introduced in Assumption 3.2, and $\gamma > 0$ is a constant that depends on the dimension of instruments $d_z$ and other tuning parameters. These constants can be taken independent of $P$ under appropriate conditions.

**Example 4.2** (Subsampled nearest neighbors)**.** Khosravi, Lewis, and Syrgkanis (2022) suggest subsampled nearest neighbors that is adaptive to the covariates' intrinsic dimension. Similarly to the previous example, the conditional mean estimator is

$$\hat{\mu}_n(z) = \sum_i \alpha_i(z)(c_n(O_i), q_n(O_i)^\top)^\top$$

where the wegiths $\alpha_i(z)$ is obtained from independently drawn $B_n$ subsamples of size $s_n$. Within each subsample, weight $1/\kappa$ is assigned to the $\kappa$-nearest neighbors of $z$. Averaging these weights over $B_n$ subsamples gives $\alpha_i(z)$.

Their Equation (8) in Theorem 1 provides a finite sample root mean squared

error bound at each fixed $z$. Under the extra conditions in Lemma C.1, we make the pointwise bound uniform[2] in $z$ and bound the first-stage error by

$$\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2 \mid \mathcal{T}_\ell] = O_P\Big(L_{\mu,n}^2 s_n^{-2/d_z} + (1+\xi_n^2)\frac{s_n}{n}\Big),$$

where $L_{\mu,n}$ is the Lipschitz constant of $\mu_{n,P}$, which may grow with $n$, and $\xi_n$ is the sieve envelope. Hence, we may take $r_{\mu,n}^2$ as the sequence in the $O_p$ term.

The bounds in Examples 4.1 and 4.2 exploit only Lipschitzness of the conditional mean of the first-stage outcomes. Faster prediction rates under stronger regularity conditions are available for other subsampling-based learners such as two-scale distributional nearest neighbors (Demirkaya et al., 2024). These results suggest first-stage learners with potentially sharper error bounds.

The last class of learners is the modern conditional distribution estimators, such as conditional diffusion and Gaussian mixture models. The theory of these estimators typically provides convergence rates under probability metrics such as total variation distance, Kullback-Leibler divergence, Wasserstein distance, and more. See Gibbs and Su (2002) for the review of such metrics. Hence, we may translate these bounds to verify the $L_2$ prediction error bound in Assumption 3.3.

**Example 4.3** (Conditional diffusion models)**.** Separate $O$ into $(V, Z)$, where $V \in \mathbb{R}^{d_V}$ and $d_V, d_Z$ are fixed. Conditional diffusion models learn the conditional distribution of $V$ given $Z$ by adding Gaussian noise to $V$ and estimating the conditional score function from the noisy version of $V$ and the conditioning variable $Z$ (Song et al., 2021; Ho and Salimans, 2022). Provided the estimated conditional distribution function, we can define

$$\widehat{\mu}_n^{(-\ell)}(z) = \int (c(v,z), q_n(v,z)^\top)^\top d\widehat{P}_n^{(-\ell)}(v \mid z),$$

and estimate the conditional means by Monte Carlo integration.

The theoretical result in Fu et al. (2024) provides a rate of convergence of the

[2]This should not be confused with the convergence in sup-norm or convergence as a function.

(integrated) total variation distance between the estimated and true conditional law. Hence, we may apply the Cauchy-Schwarz inequality and bound the prediction error by using the Hellinger distance. Then, Assumption 3.3 can be verified by

$$\mathbb{E}_P\big[\|\widehat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2 \mid \mathcal{T}_\ell\big] = \tilde{O}_P\big((1+\xi_n^2)n^{-2s/(2s+d_V+d_Z)}\big) \tag{10}$$

uniformly over $P$. Here $\tilde{O}_P(\cdot)$ refers to up to a logarithmic factor and $s$ is a smoothness parameter of the conditional density.

Gaussian mixture models provide another approach to conditional distribution estimation. Hartford et al. (2017) and Shao et al. (2025) employ mixture density networks to estimate the parameters in conditional Gaussian mixtures. Convergence rates for mixture density networks and related Gaussian mixture models are also available under several measures (Zhou et al., 2023; Ho, Yang, and Jordan, 2022; Nguyen, Nguyen, and Ho, 2023).

## 4.2 Discussion on the first-stage requirement

We briefly discuss the implications of the rate conditions on the first-stage prediction error rates. For a benchmark, we consider simple sufficient conditions that the rates in Lemma 3.1 satisfy $\tau_n^2 r_{H,n} = o(1)$ and $a_n + b_n = o(1)$. To simplify the rate conditions, suppose $\xi_n \asymp \sqrt{k_n}$, $\tau_n r_{w,n} = O(1)$, and $r_{m,n} \asymp r_{\mu,n}$. The first condition holds for many commonly used sieves. The second condition is a conservative benchmark, but imposing a faster rate for $r_{w,n}$ does not improve the sufficient rate conditions derived below. The last condition is a natural choice when no additional structure is imposed to reduce $r_{m,n}$ relative to $r_{\mu,n}$. Under these conditions, the sufficient rate requirements can be obtained as

$$\tau_n^3 k_n \sqrt{\frac{\log k_n}{n}} = o(1), \qquad r_{\mu,n} = o(\tau_n^{-1} n^{-1/4}). \tag{11}$$

The first one sets the upper bound on the growth of $k_n$. Together with the lower bound on $k_n$ implied by Assumption 3.5(ii), they set the admissible range of sieve

complexity for the statistical inference. The second one is the familiar faster-than-$n^{-1/4}$ requirement, augmented by the ill-posedness of the inverse problem.

For illustration, suppose further $L_{\mu,n} = O(1)$, which can be achieved under strong assumptions on the conditional density. Balancing the subsampling sizes in Examples 4.1 and 4.2 gives, respectively,

$$r_{\mu,n} \lesssim \left(\frac{k_n}{n}\right)^{\gamma_{\mathrm{RF}}/(1+2\gamma_{\mathrm{RF}})}, \qquad r_{\mu,n} \lesssim \left(\frac{k_n}{n}\right)^{1/(\tilde{d}_Z+2)}.$$

Since $\gamma_{\mathrm{RF}} < 1/2$ for any $d_Z$, the available GRF bound does not meet the benchmark first-stage requirement. For subsampled nearest neighbors, the bound can meet the requirement when the intrinsic dimension is one, provided $k_n$ and $\tau_n$ grows sufficiently slowly. However, we stress that these results are based on the upper bounds, whose theory often intentionally does not exploit strong regularities for more general applications. Therefore, the failure to meet the $o(n^{-1/4})$ rate of the upper bounds does not establish that these methods cannot be used in the first stage.

Other learners that exploit additional smoothness or structure have sharper prediction-error bounds that can satisfy the benchmark requirement with larger $d_Z$. For example, LASSO can exploit approximate sparsity and yield a rate that depends on the sparsity level and scales logarithmically in $d_Z$ (Belloni et al., 2012). Two-scale distributional nearest neighbors can achieve rates faster than $n^{-1/4}$ for scalar outcomes when $d_Z \leq 7$ under additional regularity conditions (Demirkaya et al., 2024). Neural networks and Mondrian forests can achieve rates faster than $n^{-1/4}$ for any fixed $d_Z$ under sufficiently strong smoothness and other regularity conditions (Schmidt-Hieber, 2020; Cattaneo, Klusowski, and Underwood, 2026). We leave verification of Assumption 3.3 for these methods to future work.

Finally, the rate requirements can be relaxed by putting additional structure on the CMR problems. In the first stage, some sieve basis functions may have natural shape restrictions. These may potentially reduce the required covering or bracketing numbers of the first-stage function class, and could result in sharper bounds. In the second stage, additional structure on the structural function, such as approximate sparsity, bounded variation, or economically motivated shape restrictions, may

reduce the rate $r_{m,n}$ relative to $r_{\mu,n}$. We leave these relaxations for future work.

# 5 Simulation Study

In this section, we study how the performance of our method changes with the number of instruments, sample size, and noise in the endogenous regressor.

**DGP1** We begin with a first-stage design that is favorable to neural networks, following the optimistic benchmark in Chen, Chen, and Tamer (2023). DGP1 has a smooth first-stage implied by a neural network. We vary the number of instruments while keeping the distribution of the underlying predictive signal fixed by aggregating the instruments. This allows us to control the instruments' dimensionality and their joint relevance separately.

We first independently generate

$$Z \sim N(0, I_{d_Z}), \quad V \sim N(0,1), \quad \epsilon \sim N(0,1).$$

Then we partition the instruments into five equally sized groups $I_r$, and define

$$S_r = \frac{\sum_{j \in I_r} Z_j}{\sqrt{|I_r|}}, \quad g(Z) = \frac{\sum_{r=1}^{5} \tanh(S_r)}{\sqrt{5\nu}},$$

where $\nu = \mathbb{E}[\tanh(N(0,1))^2] \approx 0.39$. Note $S_r \sim N(0,1)$ regardlessly of $d_Z$ and $r$. Finally, we generate the endogenous regressor and outcome as

$$\begin{aligned} X &= \Phi\left(\frac{g(Z) + \sigma_X V}{\sqrt{1+\sigma_X^2}}\right) - \frac{1}{2}, \quad U = \sigma_U\{\eta V + \sqrt{1-\eta^2}\epsilon\}, \\ Y &= \theta_0(X) + U = 2\sin(\pi X) + U. \end{aligned}$$

There are three DGP parameters $(\sigma_X, \sigma_U, \eta)$. The parameter $\sigma_X$ controls the amount of variation in the endogenous regressor that is not explained by the instruments. The parameter $\sigma_U$ determines the standard deviation of the structural

error, while $\eta$ controls its dependence on the shared shock $V$ and hence the degree of endogeneity. By construction, $\mathrm{Corr}(U, V) = \eta$.

**First-stage learners** We approximate $\theta_0$ with quadratic and cubic B-splines having three or five interior knots. The first-stage jointly predicts the conditional mean of the outcome and spline basis functions. We use the generalized random forests (GRF; Athey, Tibshirani, and Wager, 2019) and feedforward neural networks as first-stage learners. We do 5-fold cross-fitting.

For GRF, we fix the subsampling proportion as 0.4 in all simulations[3]. We tried different subsampling sizes in pilot simulations and found that GRF in general works better with a larger subsampling size in our first-stage. We fix other tuning parameters, such as minimum split portion and variable selections, to the default in the authors' R package.

For the neural networks, we consider an architecture with three hidden layers with sixteen neurons each. We use the ReLU activation function and Adam as a training algorithm with a batch size of 128. We try different learning rates $\{0.001, 0.003, 0.01\}$ and $L_2$ regularization parameters $\{0.001, 0.01\}$ in the Python package scikit-learn. For each DGP configuration, we generate three extra datasets and decide the tuning parameters by 5-fold out-of-sample prediction. In actual Monte Carlo simulations, we fixed the chosen tuning parameters.

For the conditional diffusion model, we separately estimate the conditional distribution of $Y$ and $X$ given $Z$. For $X$, we apply a logit transformation on $X + 1/2$ to make its support a real line. We use three hidden layers of 32 neurons and the SiLu activation function. We use the Adam algorithm with learning rate 0.001 and a batch size of 256. In each training fold, we reserved 20% of the training data for validation of early stopping. After fitting the model conditional distribution, we sampled 1024 Monte Carlo draws for numerical integration.

[3]Theoretically, the subsampling size needs to satisfy $s_n/n \to 0$, so the proportion has to decrease, but we fix it at 0.4 due to computational burden. More importantly, as discussed in Section 4.2, the use of GRF is not theoretically justified. However, we found it still works well with a small number of instruments (less than 5) in our simulations. The results need to be interpreted with caution.

**Simulation settings** We do 200 Monte Carlo repetitions for neural networks and 50 repetitions for GRF due to computational burden. The Monte Carlo repetitions of the diffusion model are currently 50. We set as baseline $\sigma_X = \sqrt{0.1} \approx 0.31$, $\sigma_U = 0.5$, and $\eta = 0.9$. In the first experiment, we set the DGP parameters to the baseline and vary the number of instruments $d_Z \in \{5, 10, 20\}$ and the sample size $n \in \{2500, 5000, 10000\}$. In the second experiment, we fix $n = 5000$, $\sigma_U = 0.5$, $\eta = 0.9$, and vary $\sigma_X \in \{0.2, 0.31, 0.5\}$ and $d_Z \in \{5, 10, 20\}$.

In the baseline design, $\text{Corr}(X, U) \approx 0.26$. To quantify instrument relevance, we define $R_X^2 := \text{Var}(\mathbb{E}[X|Z])/\text{Var}(X)$, which measures the proportion of variation in $X$ explained by the whole instruments jointly. We note that the conditional mean $\mathbb{E}[X|Z]$ is for reference (useful in linear IV), and it is not the object directly used in our first-stage[4]. In the baseline design, $R_X^2$ is approximately 90.4%, which is very high in empirical settings. In the second experiment with noise, $R_X^2 \approx 0.96$ when $\sigma_X = 0.2$ and $R_X^2 \approx 0.79$ when $\sigma_X = 0.5$. We set it deliberately high because we find that the finite sample performance of our inference deteriorates substantially with the noise in instruments.

We perform two Wald inferences with nominal level 5%. One is the point evaluation at $x = 0$, and the other is the average derivative. For point evaluation, the null value $\phi_0$ is 0. For average derivative, we trim the top and bottom 5 percentiles of the observations, and test for $\mathbb{E}[\theta_0'(X)|X_{(0.05)} \leq X \leq X_{0.95}]$ where $X_{(\alpha)}$ refers to $\alpha$-quantile of $X$. The null value is fixed by Monte Carlo at $\phi_0 = 4.34$. As a performance report, we provide local power curves. To obtain the curve, we fix the DGP and instead shift the null value by $\phi_0 + \kappa\sqrt{V_n/n}$ for $\kappa \in \{0, 0.1, \cdots, 4\}$. $V_n$ is fixed from separate Monte Carlo simulations. At $\kappa = 0$, the rejection frequency indicates the (empirical) size of the test, so we do not separately report size.

**Results** Figure 1 and 2 respectively illustrate the local power of the point evaluation and average derivative Wald test by sample size and instrument dimension.

[4]For instance, Bruns-Smith (2025) reports $R_Y^2$ as a first-stage performance metric because the method uses a reduced form prediction of $Y$ given $Z$. For the conditional mean of $Y$ and the quadratic B-spline (with three knots) basis functions, the analogue measures are $R_Y^2 = 0.69$ and $R_p^2 = (0.82, 0.73, 0.62, 0.62, 0.73, 0.82)$, respectively, in the baseline design.

The reported second-stage sieve is a quadratic B-spline with three interior knots. In both tests and all configurations, the size was kept close to the nominal level of 5%. In Figure 1, when the sample size was small ($n = 2500$), the performance decreased substantially as the number of instruments grew. This is expected because modern machine learning methods can accommodate many variables computationally, but they need not eliminate the exponential growth in required sample sizes. Nevertheless, the power recovered fast in our simulations as the sample size gets larger. For the neural net with $d_Z = 20$, the rejection frequency at $\kappa = 4$ increased from 30% to 71% to 93% as $n$ increases. The corresponding average derivative rejection rates were 50%, 98%, and 99%. The average derivative testing achieved nearly ideal rejection rates in all scenarios when $n = 10000$.

Although the upper bound in Example 4.1 does not deliver a first-stage convergence rate faster than $n^{-1/4}$, we find that GRF performs well when the number of instruments is small. Its performance is less sensitive to tuning parameters, and it outperforms neural networks in some low-dimensional configurations. However, GRF performance deteriorates rapidly as the number of instruments increases, whereas neural networks perform better with moderately many instruments.

Figure 3 examines the sensitivity of local power to the noise level $\sigma_X$, holding the sample size ($n = 5000$) and other DGP parameters fixed. The results show that increased noise in the endogenous regressor can substantially reduce rejection frequencies, particularly when the number of instruments is large. The rejection frequencies of the neureal net at $\kappa = 4$ were 99%, 99%, 45% for $\sigma_X = 0.20, 0.31, 0.50$, resepectively.

We may expect that greater noise in the endogenous regressor may affect the performance in multiple channels: aggravating population (sieve) ill-posedness and making the first-stage regressions more challenging by reducing their signal-to-noise ratio. These effects can reinforce each other, as weaker population curvature amplifies first-stage estimation errors and increases the risk of instability in the estimated Hessian. In DGP1, increasing the number of instruments does not change population sieve ill-posedness, but makes the conditional means harder to estimate. The bottom-right panel of Figure 3 suggests this potential vulnerability to noise could

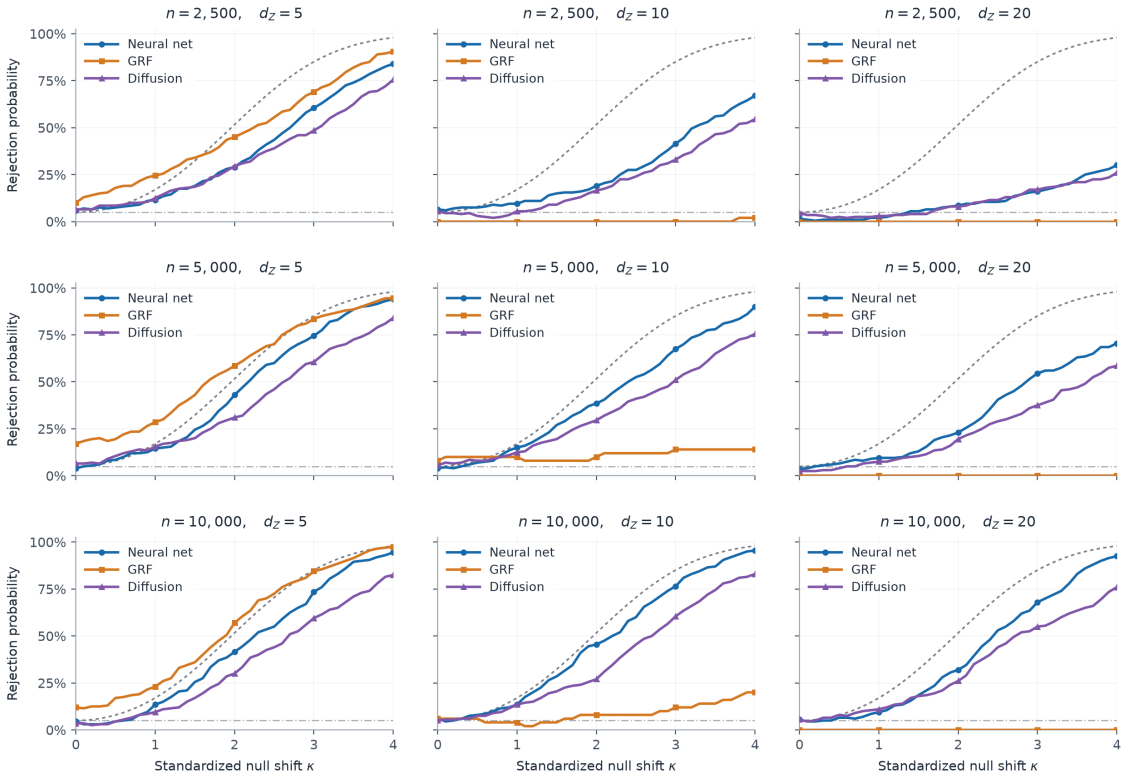



Figure 1: Local power of point evaluation Wald test by sample size $n$ and the instrument dimension $d_Z$. Nominal significance level is 5%. The dotted curve represents the Gaussian reference rejection probability, and the horizontal line indicates the nominal significance level.

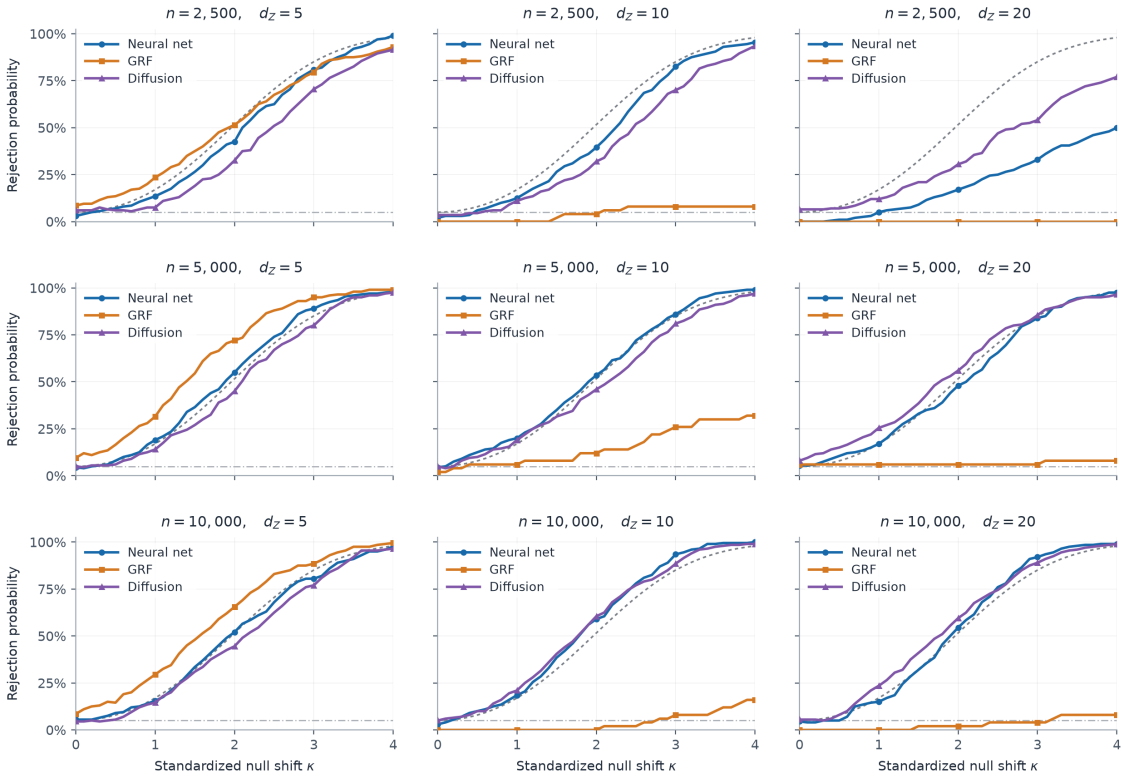



Figure 2: Local power of average derivative Wald test by sample size $n$ and the instrument dimension $d_Z$. Nominal significance level is 5%. The dotted curve represents the Gaussian reference rejection probability, and the horizontal line indicates the nominal significance level.

be pronounced in high-dimensional settings. The rejection rate of the neural net was 26% when $\kappa = 4$ in the plot. Also, while we find neural networks are highly sensitive to tuning, we only tried a few primitive tunings. Hence, the performance could improve with finer tuning.

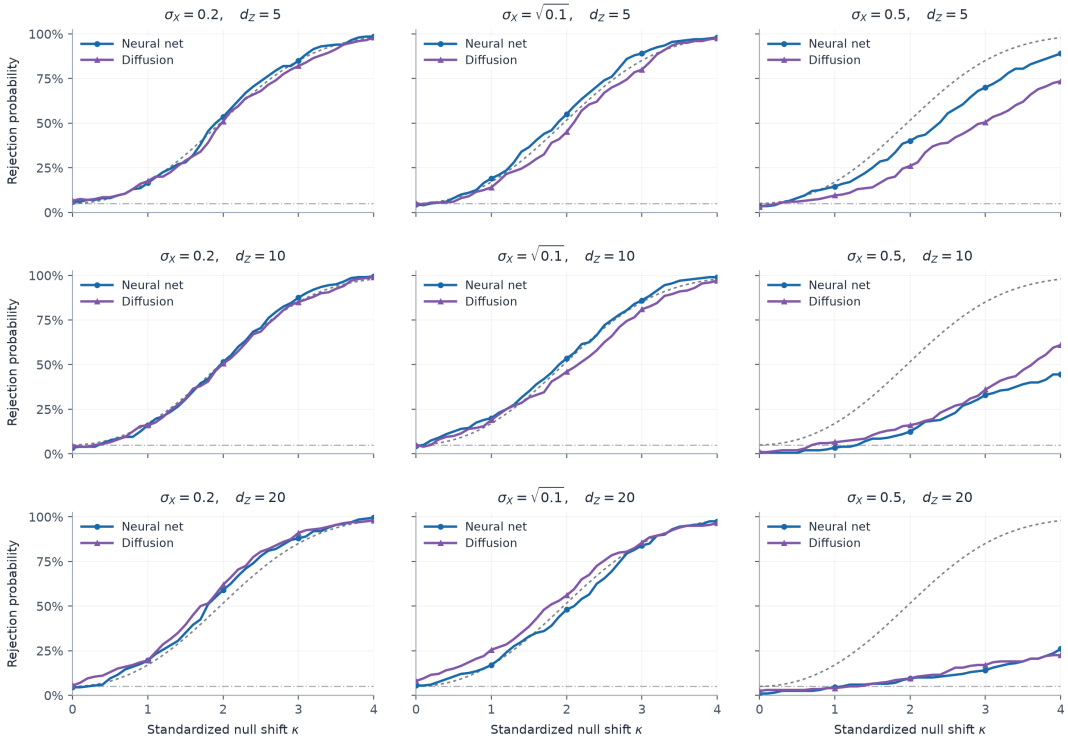



Figure 3: Local power of average derivative Wald test by noise $\sigma_X$ and the instrument dimension $d_Z$. Nominal significance level is 5%. The dotted curve represents the Gaussian reference rejection probability, and the horizontal line indicates the nominal significance level.

# 6 Empirical Application

In this section, we revisit the empirical work of Beegle, De Weerdt, and Dercon (2011), who studied migration and economic mobility in Tanzania. The authors investigate whether geographic mobility improves individuals' living standards with the Kagera Health and Development Survey data, which tracks individuals from

their original households in northwestern Tanzania between 1991 and 2004. While the original paper examines several aspects of migration and economic mobility, we focus on one of their instrumental variable regression, which estimates the effect of migration distance on consumption growth. We extend the linear IV specification by nonparametrically estimating the effect of migration distance on consumption growth.

Particularly, we consider the 2SLS in Table 9, column 4 of their study. Let $Y_i$ denote the change in log real consumption per capita between 1991 and 2004, and let $X_i = \log(1 + D_i)$, where $D_i$ is migration distance in kilometers. The original 2SLS model considers

$$Y_i = \beta X_i + C_i^\top \gamma + \alpha_{H_i} + U_i,$$

where $C_i$ contains twenty eight individual and family characteristics controls and $\alpha_{H_i}$ is an origin-household fixed effect. The six (excluded) instruments are household-position indicators and interactions involving age rank, distance to the regional capital, and the original rainfall-shock measure. The data contain 3227 individuals and 681 origin households. Of these individuals, 2147 have zero recorded migration distance, and 1080 have positive distance. In the 2SLS regression, the authors find the slope of $X$ is 0.104 with strong significance.

To allow for a nonlinear relationship between migration distance and consumption growth, we consider the following partially linear NPIV specification while retaining the original controls, origin-household fixed effects, and excluded instruments:

$$Y_i = h(X_i) + C_i^\top \gamma + \alpha_{H_i} + U_i, \qquad \mathbb{E}[U_i \mid Z_i, C_i, H_i] = 0.$$

We approximate $h$ with quadratic B-splines with five equally spaced knots. Without loss of generality, we normalize $h(0) = 0$ by subtracting its estimated value at zero and absorbing the differences into the intercept in $C$. Hence, the redefined function $h(x) = h^{raw}(x) - h(0)$ measures the effect of migration distance relative to staying in the original location. In the first-stage, we employ GRF and neural networks. For

GRF, we set the subsampling proportion to 0.4 and retain all other tuning parameters as the R package default. For neural networks, we use three layers of sixteen nodes with ReLU activation function and the Adam optimizer[5]. The first-stage inputs include the excluded instruments, original controls, and origin-household indicators. We do 5-fold cross-fitting[6].

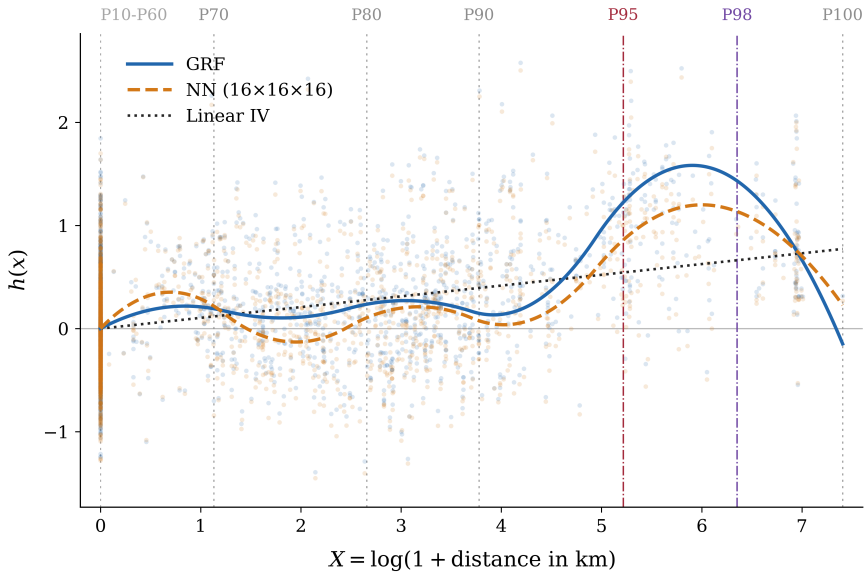



Figure 4: Consumption-growth curves against $X = \log(1 + \text{distance in km})$. The B-spline curves use quadratic B-splines with five interior knots. The dotted line is the original linear IV fit. All curves are normalized to $h(0) = 0$. The scatter points show residualized outcomes, with colors matching the corresponding fitted curves. Vertical lines mark deciles (gray), P95 (red), and P98 (purple).

Figure 4 presents the estimated structural functions alongside the linear IV estimate. Both GRF and neural network estimates suggest that the relationship between

[5]We set the learning rate and $L_2$ regularization to 0.001. We used a batch size of 128 and used 500 training epochs. We set aside 15% of the training samples as validation samples and use early stopping with patience of 25 epochs and tolerance of $10^{-4}$. All training ended before reaching the epoch limit.

[6]The individuals that share the same household can be related. We tried several cross-fittings with and without preserving household relationships, but the results were similar. We report the one that considers household relationships in both origin (year 1991) and destination (year 2004) households.

| | Estimate | SE | Estimate | SE |
|---|---|---|---|---|
| Linear IV slope | 0.104** | 0.043 | - | - |
| | GRF | | NN | |
| Average derivative, P70-90 | -0.026 | (0.064) | -0.053 | (0.050) |
| Average derivative, P90-95 | 0.582*** | (0.112) | 0.396*** | (0.104) |
| Average derivative, P90-98 | 0.564*** | (0.099) | 0.434*** | (0.075) |
| Difference (P90-95)-(P70-90) | 0.607*** | (0.142) | 0.450*** | (0.124) |

Table 1: Migration distance and consumption growth in Tanzania. The table reports the linear IV estimate and average derivatives of the estimated structural function over selected percentile ranges of migration distance. GRF and NN denote the first-stage learners. Standard errors are reported in the adjacent columns. P70–90 denotes the 70th to 90th percentiles of the transformed migration distance. ** and *** indicate significance at the 5% and 1% levels, respectively.

migration distance and consumption growth is relatively flat over most of the distribution, but increases sharply at higher migration distances. We therefore compare average derivatives across different ranges of migration distance to investigate potential heterogeneity in their marginal effect.

Before presenting the inference results, we note three caveats. First, our current theory assumes iid samples, so our standard errors do not account for the clustered error within households. We leave this generalization to future work. Second, the number of households increases with sample size. Although our theory does not preclude an increasing number of first-stage inputs, the theory for employed machine learning methods may not allow it. In particular, the theory underlying GRF assumes a fixed number of inputs, although it can be relatively large. Finally, around two-thirds of observations are concentrated at the zero boundary of migration distance, while nonparametric methods sometimes have unstable behaviors near the boundary. This concentration may affect the stability of the sieve estimation. Also, the interesting findings are concentrated near the right boundary.

Table 1 provides suggestive evidence of heterogeneity. Both GRF and NN estimates show that the average derivative is close to zero between the 70th and 90th percentiles but significantly larger between the 90th and 95th percentiles. The differ-

ences between the average derivatives are also statistically significant. Nevertheless, we interpret these results as exploratory for the aforementioned limitations in the inference. De Weerdt, Christiaensen, and Kanbur (2021) also find larger consumption gains among migrants to distant cities than among those moving to nearby towns in Tanzania by distinguishing migrant destinations in the analysis.

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# A Gaussian coupling of the sieve coefficients

In this section, we provide a Gaussian coupling of the linear representation in Lemma 3.1. The coupling can be used as a building block for various nonstandard inference problems.

**Assumption A.1.** (i) Assumptions 3.3 and Assumption 3.4(iii) hold uniformly over $P \in \mathcal{P}$, with constants and deterministic rate sequences independent of $P$. (ii) Suppose there is a deterministic sequence $\tau_n^2$ such that $\sup_P \|G_{n,P}^{1/2} H_{n,P}^{-1/2}\|_{op} \leq \tau_n$ and $\tau_n^2 r_{H,n} = o(1)$. (iii) For a given deterministic sequence $D_{n,P}$ for each $P \in \mathbf{P}$, $\sup_{P\in\mathcal{P}} \mathbb{E}_P\left[\left\|D_{n,P} H_{n,P}^{-1} \psi_{n,P}(O)\right\|_2^3\right] < \infty$. (iv) Define $\Delta_{D,P} := n^{-1/2}\mathbb{E}_P[\|D_{n,P} H_{n,P}^{-1}\psi_{n,P}(O)\|_2^3]$. For a deterministic sequence $c_n \to 0$, suppose $c_n^{-3} k_n \sup_P \Delta_{D,P} \to 0$ and $(a_n + b_n)\sup_P \|D_{n,P}\|_{op} = o(c_n)$.

$D_{n,P}$ is a problem specific choice. For example, $D_{n,P}$ can be taken as $n^{-1/2} r_n I_{k_n}$ to verify Assumption 3.2(i) in Fang and Seo (2021), or can be taken as $(H_{n,P}^{-1}\Omega_{n,P} H_{n,P}^{-1})^{-1/2}$ to verfify Condition C in Chernozhukov, Lee, and Rosen (2013). (i) For Assumption 3.3, the examples provided yield bounds with the constants that do not depend on $P$. Requiring Assumption 3.4(iii) uniformly over $P$ restricts the class of DGP so that the sieve approximation error measured in weak norm is bounded by a common deterministic rate $r_{w,n}$. Thus, the rate is chosen to accommodate the largest weak norm approximation error over $P \in \mathcal{P}$. (ii) imposes an upper envelope on the sieve measure of ill-posedness uniformly over $P$ (Chen and Christensen, 2018). (iii) is the third momoment condition for Yurinskii's coupling. (iv) $c_n$ is a tolerance of approximation a researcher chose. For instance, it can be taken as $\log(n)$ or $n^{-a}$ for some $a > 0$.

**Lemma A.1.** Suppose Assumption A.1 holds. For a deterministic sequence of matrix $D_{n,P}$, there exists Gaussian random vectors $W_{n,P}^D \sim N(0, D_{n,P}\Sigma_{n,P} D_{n,P}^\top)$ such that

$$\|\sqrt{n} D_{n,P}(\hat{\beta}_n - \beta_{P,n}) - W_{n,P}^D\|_2 = o_P(c_n)$$

uniformly over $P \in \mathbf{P}$.

The proof is a direct application of the Yurinskii coupling (Yurinskii, 1977).

# B Proofs

## B.1 Proofs for Section 3

To relieve notational burden, we write $f_i = f(A_i)$ for any function $f$ and random variable $A_i$.

**Lemma B.1** (Hessian estimation error)**.** Suppose Assumptions 3.1(i)(iv)(vi), 3.2(i)(ii), 3.3, and 3.4(iv) hold.

$$\|\hat{H}_n - H_n\|_{op} = O_p(\xi_n\sqrt{\frac{\log k_n}{n}} + \frac{r_{\mu,n}\xi_n}{\sqrt{n}} + r_{\mu,n}^2).$$

*Proof of Lemma B.1.* Denote $\tilde{H}_n := \mathbb{E}_n[\sigma_i^{-2}\{\mu_{qi}q_i^\top + (q_i - \mu_{qi})\mu_{qi}^\top\}]$ and $e(Z_i) := \hat{\mu}_q^{(-\ell)}(Z_i) - \mu_q(Z_i)$ for $i \in I_\ell$.

$$\begin{aligned}\hat{H}_n &\equiv \mathbb{E}_n\Big[\frac{1}{\sigma^2}\Big\{\hat{\mu}_{qi}q_i^\top + (q_i - \hat{\mu}_{qi})\hat{\mu}_{qi}^\top\Big\}\Big] \\ &= \tilde{H}_n + \mathbb{E}_n\Big[\frac{1}{\sigma^2}\Big\{e(q-\mu)^\top + (q-\mu)e^\top - ee^\top\Big\}\Big].\end{aligned}$$

By Assumption 3.4(iv), $\mathbb{E}_P[\|e(Z_i)\|_2^2|\mathcal{T}] = O_p(r_{\mu,n}^2)$ where $\mathcal{T}$ denotes the training fold(s). Meanwhile, by Assumption 3.1(iv) and Assumption 3.2(ii), $\|q_n(O)\|_2 \precsim \xi_n$ $P$-a.s.. By Jensen's inequality, $\|\mu_q(Z)\|_2 \precsim \xi_n$ $P$-a.s., hence, by triangle inequality $\|q - \mu_q\|_2 \precsim \xi_n$ $P$-a.s.. Together with Assumption 3.4(iv), in each fold $\ell$,

$$\begin{aligned}\mathbb{E}\Big[\Big\|\frac{1}{n_\ell}\sum_{i\in I_\ell}[e_i(q_i-\mu_i)^\top/\sigma_i^2]\Big\|_{op}^2\Big|\mathcal{T}_\ell\Big] &\leq \mathbb{E}\Big[\Big\|\frac{1}{n_\ell}\sum_{i\in I_\ell}[e_i(q_i-\mu_i)^\top/\sigma_i^2]\Big\|_F^2\Big|\mathcal{T}_\ell\Big] \\ &= \frac{1}{n_\ell}\mathbb{E}\Big[\frac{\|e_i(q_i-\mu_i)^\top\|_F^2}{\sigma_i^4}\Big|\mathcal{T}_\ell\Big] \\ &= \frac{1}{n_\ell}\mathbb{E}\Big[\frac{\|e_i\|_2^2\|q_i-\mu_i\|_2^2}{\sigma_i^4}\Big|\mathcal{T}_\ell\Big] \\ &\precsim O_p\big(\frac{r_{\mu,n}^2\xi_n^2}{n_\ell}\big) = O_p\big(\frac{r_{\mu,n}^2\xi_n^2}{n}\big).\end{aligned}$$

Aggregating across $\ell$, by the law of iterative expectation and the Markov inequality,

$\left\|\mathbb{E}_n\left[\frac{e_i(q_i-\mu_i)^\top}{\sigma_i^2}\right]\right\|_{op} = O_p(\frac{r_{\mu,n}\xi_n}{\sqrt{n}})$. The second remainder has the same bound. Similarly, the last remainder term is $O_p(r_{\mu,n}^2)$ by Assumption 3.4(iv), the Cauchy-Schwarz inequality, and the Markov inequality. Hence,

$$\|\hat{H}_n - \tilde{H}_n\|_{op} = O_p(\frac{r_{\mu,n}\xi_n}{\sqrt{n}} + r_{\mu,n}^2).$$

Notice $\tilde{H}_n = \mathbb{E}_n[\frac{1}{\sigma^2}(q_i q_i^\top - (q_i - \mu_{qi})(q_i - \mu_{qi})^\top)]$. We may apply Lemma 6.2 of Belloni et al. (2015) separately to each term. From the bounds on $\|q_i\|_2$ and $\|q_i - \mu_{qi}\|_2$ above, the envelope condition holds with $M = \xi_n^2/\underline{\sigma}^2$ and $M = 4\xi_n^2/\underline{\sigma}^2$, respectively. Then, provided $\xi_n\sqrt{\frac{\log k_n}{n}} = o(1)$,

$$\|\tilde{H}_n - H_n\|_{op} = O_p\Big(\frac{\xi_n^2 \log k_n}{n} + \xi_n\sqrt{\frac{\log k_n}{n}}\Big) = O_p\Big(\xi_n\sqrt{\frac{\log k_n}{n}}\Big).$$

Finally, by the triangle inequality,

$$\|\hat{H}_n - H_n\|_{op} = O_p\Big(\xi_n\sqrt{\frac{\log k_n}{n}} + \frac{r_{\mu,n}\xi_n}{\sqrt{n}} + r_{\mu,n}^2\Big).$$

□

Lemma B.2 is a standalone lemma that collects several established results in the literature. See, for instance, Appendix F.1 of Chen and Christensen (2018) and Appendix A.1 of Belloni et al. (2015). A proof is provided for completeness.

**Lemma B.2.** Let $H_n$ be a deterministic symmetric positive-definite $k_n \times k_n$ matrix. Suppose $\lambda_{\min}(H_n) \geq c_n > 0$ for some deterministic sequence $c_n > 0$ that can go to 0. Let $\hat{H}_n$ be an estimator of $H_n$, which is symmetric and $k_n \times k_n$. Define $\Delta_{H,n} := H_n^{-1/2}(\hat{H}_n - H_n)H_n^{-1/2}$ and suppose that $\|\Delta_{H,n}\|_{op} = o_p(1)$ and $\xi_n\sqrt{\frac{\log k_n}{n}} = o(1)$. Then, $\hat{H}_n$ is positive definite (and hence invertible[7]) with probability approaching to 1. Moreover, $\|\hat{H}_n^{-1}\|_{op} = O_p(c_n^{-1})$ and $\|\hat{H}_n^{-1} - H_n^{-1}\|_{op} = O_p(\|\hat{H}_n - H_n\|_{op}/c_n^2)$.

[7]Define $\hat{H}_n^{-1}$ as any matrix, for instance 0 matrix, if $\hat{H}_n$ is singular.

The second half of the result might not be useful, if we take care of the product terms separately as in Fang and Seo (2021).

*Proof of Lemma B.2.* By definition of $\Delta_{H,n}$, $\hat{H}_n = H_n^{1/2}(I_{k_n} + \Delta_{H,n})H_n^{1/2}$. Fix $\epsilon > 0$. On the event $\mathcal{A}_n(\epsilon) = \{\|\Delta_{H,n}\|_{op} \leq \epsilon\}$, for any nonzero $u \in \mathbf{R}^{k_n}$, let $v := H_n^{1/2}u$. Then,

$$\begin{aligned} u^\top \hat{H}_n u &= u^\top H_n^{1/2}(I_{k_n} + \Delta_{H,n})H_n^{1/2}u \\ &= \|v\|_2^2 + v^\top \Delta_{H,n} v \\ &\geq \|v\|_2^2 - \|\Delta_{H,n}\|_{op}\|v\|_2^2 \\ &\geq (1-\epsilon)\|v\|_2^2 \\ &= (1-\epsilon)u^\top H_n u \\ &> 0. \end{aligned}$$

By assumption, $Pr(\mathcal{A}_n(\epsilon)) \to 1$. Hence, $Pr(\hat{H}_n > 0) \geq Pr(\mathcal{A}_n(\epsilon)) \to 1$.

Define $\hat{H}_n^{-1} = H_n^{-1/2}(I_{k_n} + \Delta_{H,n})^{-1}H_n^{-1/2}$. For the same reason, $(I_{k_n} + \Delta_{H,n})$ is positive definite on $\mathcal{A}_n(\epsilon)$ and hence invertible. By construction, $\hat{H}_n^{-1}$ is the inverse of $\hat{H}_n$. Again, on the event $A_n(\epsilon)$, from the definition of eigenvalue,

$$\|\hat{H}_n^{-1}\|_{op} = \frac{1}{\lambda_{\min}(\hat{H}_n)} \leq \frac{1}{(1-\epsilon)\lambda_{\min}(H_n)} \leq \frac{1}{(1-\epsilon)c_n}. \tag{12}$$

The last inequality follows from $\|H_n^{-1}\|_{op} = 1/\lambda_{\min}(H_n) \leq 1/c_n$.

Finally, on the same event, both inverses exists, and

$$\begin{aligned} \|\hat{H}_n^{-1} - H_n^{-1}\|_{op} &= \|\hat{H}_n^{-1}(H_n - \hat{H}_n)H_n^{-1}\|_{op} \\ &\leq \|\hat{H}_n^{-1}\|_{op}\|\hat{H}_n - H_n\|_{op}\|H_n^{-1}\|_{op} \\ &\leq \frac{(1-\epsilon)^{-1}\|\hat{H}_n - H_n\|_{op}}{c_n^2} \\ &= O_p(\frac{\|\hat{H}_n - H_n\|_{op}}{c_n^2}). \end{aligned}$$

□

*Proof of Lemma 3.1 (ALR).* Define $\tilde{\Psi}_n = \mathbb{E}_n[\psi_i(\beta_{P,n})]$ where $\psi_i$ is defined in Equation (4). Recall we assume $\beta_{P,n}$ is an interior solution. We may decompose as

$$\begin{aligned}\sqrt{n}(\hat{\beta}_n - \beta_{P,n}) &= -H_n^{-1}\frac{1}{\sqrt{n}}\sum_i \psi_i + (H_n^{-1} - \hat{H}_n^{-1})\frac{1}{\sqrt{n}}\sum_i \psi_i - \hat{H}_n^{-1}\sqrt{n}(\hat{\Psi}_n - \tilde{\Psi}_n) \\ &:= -H_n^{-1}\mathbb{G}_n\psi_i + R_{H,n} + R_{\psi,n} \qquad (13)\end{aligned}$$

The first term is the leading term in the lemma. It is centered due to the interior solution and the first order condition.

For the first remainder $R_{H,n}$, we first bound $\|H_n^{-1}\mathbb{G}_n\psi_i\|_2$. Recall $\psi_i = (\rho_i - m_i)\mu_{qi}/\sigma_i^2 + m_i q_i/\sigma_i^2$. The first term is by independence, LIE, Assumption 3.1(ii), Assumption 3.4(iii),

$$\begin{aligned}\mathbb{E}_P\Big[\Big\|H_n^{-1}\frac{1}{\sqrt{n}}\sum_i \frac{(\rho_i - m_i)\mu_{qi}}{\sigma_i^2}\Big\|_2^2\Big] &\precsim \mathbb{E}_P[\frac{\mu_q^T H_n^{-2}\mu_q}{\sigma_i^2}] \\ &= \mathrm{tr}(H_n^{-1}E_P[\frac{\mu_{qi}\mu_{qi}^\top}{\sigma_i^2}]H_n^{-1}) \\ &= \mathrm{tr}(H_n^{-1}) \precsim \tau_n^2 k_n. \qquad (14)\end{aligned}$$

The second term is by independence, the Cauchy-Schwarz inequality, Assumption 3.4(iii),

$$\begin{aligned}\mathbb{E}_P\Big[\|H_n^{-1}\frac{1}{\sqrt{n}}\sum_i \frac{m_i q_i}{\sigma_i^2}\|_2^2\Big] &= \mathbb{E}_P[\frac{m_i^2}{\sigma_i^4}q_i^\top H_n^{-2}q_i] \\ &\leq \|H_n^{-1}\|_{op}^2 E_P[\frac{m_i^2\|q_i\|_2^2}{\sigma_i^4}] \precsim \tau_n^4\xi_n^2 r_{w,n}^2. \qquad (15)\end{aligned}$$

The last inequality holds from $\mathbb{E}_P[m_i^2/\sigma_i^2] = Q_P(\theta_{P,n}) \leq Q_P(\Pi_n\theta_P) = O(r_{w,n}^2)$.

Hence, by Equations (14), (15), and Lemma B.1,

$$\begin{aligned}
\|R_{H,n}\|_2 &= \Big\|\hat{H}_n^{-1}(\hat{H}_n - H_n)H_n^{-1}\frac{1}{\sqrt{n}}\sum_i \psi_i\Big\|_2 \\
&\leq \|\hat{H}_n^{-1}\|_{op}\|(\hat{H}_n - H_n)\|_{op}\Big\|H_n^{-1}\frac{1}{\sqrt{n}}\sum_i \psi_i\Big\|_2 \\
&= O_p(\tau_n^2\{\xi_n\sqrt{\frac{\log k_n}{n}} + \frac{r_{\mu,n}\xi_n}{\sqrt{n}} + r_{\mu,n}^2\}\{\tau_n\sqrt{k_n} + \tau_n^2\xi_n r_{w,n}\}). \qquad (16)
\end{aligned}$$

For the second remainder $R_{\psi,n}$, we first derive the estimation error of the $\hat{\Psi}_n$ term. Denoting $e_\mu = \hat{\mu}_q - \mu_q$ and $e_m = \hat{m}(\cdot, \beta_{P,n}) - m(\cdot, \beta_{P,n})$,

$$\begin{aligned}
\hat{\Psi}_n(\beta_{P,n}) &:= \frac{1}{n}\sum_i \frac{1}{\sigma_i^2}\Big\{\rho_i\hat{\mu}_i + \hat{m}_i(q_i - \hat{\mu}_i)\Big\} \\
&= \frac{1}{n}\sum_i \frac{1}{\sigma^2}\Big\{\rho(\mu + e_\mu) + (m + e_m)(q - \mu - e_\mu)\Big\} \\
&= \tilde{\Psi}_n(\beta_{P,n}) + \frac{1}{n}\sum_i \frac{1}{\sigma^2}\Big\{(\rho - m)e_\mu + e_m(q - \mu) - e_m e_\mu\Big\}
\end{aligned}$$

Similarly to the proof of Lemma B.1, in each fold $\ell$, by indendency, Assumption 3.1(ii), the Cauchy-Schwarz inequality, and Assumption 3.4(iv), 3.3,

$$\begin{aligned}
\mathbb{E}\Big[\Big\|\frac{1}{n_\ell}\sum_{i\in I_\ell}[(\rho - m)e_\mu/\sigma^2]\Big\|_2^2\Big|\mathcal{T}_\ell\Big] &= \frac{1}{n_\ell}\mathbb{E}[\sigma_i^{-4}\|(\rho_i - m_i)e_{\mu,i}\|_2^2|\mathcal{T}_\ell] \\
&\precsim \frac{1}{n_\ell}\mathbb{E}\big[\|e_{\mu,i}\|_2^2|\mathcal{T}_\ell\big] = O_p(\frac{r_{\mu,n}^2}{n})
\end{aligned}$$

Similarly,

$$\mathbb{E}\Big[\Big\|\frac{1}{n}\sum_i[e_m(q - \mu)/\sigma^2]\Big\|_2^2\Big|\mathcal{T}_\ell\Big] \leq \frac{1}{n_\ell}\mathbb{E}\Big[\frac{e_m^2\|q - \mu_q\|_2^2}{\sigma^4}\Big|\mathcal{T}_\ell\Big] = O_p(\frac{\xi_n^2 r_{m,n}^2}{n})$$

The last $e_m e_\mu$ term is by the Cauchy-Schwarz and Markov inequalities, $O_p(r_{\mu,n}r_{m,n})$.

Hence,

$$\|\hat{\Psi}_n(\beta_{P,n}) - \tilde{\Psi}_n(\beta_{P,n})\|_2 = O_p(\frac{r_{\mu,n}}{\sqrt{n}} + \frac{\xi_n r_{m,n}}{\sqrt{n}} + r_{m,n} r_{\mu,n}). \tag{17}$$

Combining Equation (13), Lemma B.2, (17)

$$\begin{aligned} \|R_{\psi,n}\|_2 &\leq \sqrt{n}\|\hat{H}_n^{-1}\|_{op}\|\hat{\Psi}_n - \tilde{\Psi}_n\|_2 \\ &= O_p(\tau_n^2 r_{\mu,n} + \xi_n r_{m,n} + \sqrt{n} r_{m,n} r_{\mu,n}) = O_p(b_n). \end{aligned} \tag{18}$$

Finally, define $R_n = R_{H,n} + R_{\psi,n}$, then

$$\|R_n\|_2 \leq \|R_{H,n}\|_2 + \|R_{\psi,n}\|_2 = O_p(a_n + b_n).$$

□

*Proof of Theorem 3.1.* Suppose $V_n$ is known. By Assumption 3.5 and 3.6(ii),

$$\begin{aligned} \frac{\sqrt{n}}{\sqrt{V_n}}(\phi(\hat{\theta}_n) - \phi(\theta_P)) &= \frac{\sqrt{n}}{\sqrt{V_n}}(\phi(\hat{\theta}_n) - \phi(\theta_{P,n})) + \frac{\sqrt{n}}{\sqrt{V_n}}(\phi(\theta_{P,n}) - \phi(\theta_P)) \\ &= \frac{\sqrt{n}}{\sqrt{V}_n} l_n^\top (\hat{\beta}_n - \beta_{P,n}) + o_p(1). \end{aligned}$$

By Lemma 3.1,

$$\begin{aligned} \frac{\sqrt{n}}{\sqrt{V}_n} l_n^\top (\hat{\beta}_n - \beta_{P,n}) &= -\frac{1}{\sqrt{V}_n} l_n^\top \frac{1}{\sqrt{n}} \sum_i \psi_{ni} + \frac{1}{\sqrt{V}_n} l_n^\top (R_{H,n} + R_{\psi,n}) \\ &= -\frac{1}{\sqrt{n}} \sum_i \frac{S_{ni}}{\sqrt{V_n}} + \frac{1}{\sqrt{V}_n} l_n^\top (R_{H,n} + R_{\psi,n}). \end{aligned}$$

By the triangle inequality, the Cauchy-Schwarz inequality, and the assumption that $\|l_n\|_2(a_n + b_n)/\sqrt{V_n} = o_p(1)$,

$$\Big|\frac{1}{\sqrt{V_n}}(l_n^\top R_{H,n} + l_n^\top R_{\psi,n})\Big| \leq \frac{\|l_n\|_2}{\sqrt{V_n}}(\|R_{H,n}\|_2 + \|R_{\psi,n}\|_2) = O_p(\frac{\|l_n\|_2(a_n + b_n)}{\sqrt{V_n}}) = o_p(1).$$

Define $X_{ni} = \frac{S_{ni}}{\sqrt{V_n}}$ and apply the Lindeberg-Feller CLT with Assumption 3.6. Then, conclude by Lemma B.3 and Slutsky's lemma. □

**Lemma B.3** (Wald variance estimator)**.** Suppose $\tau_n^2 r_{H,n} = o(1)$ and $\|l_n\|_2(a_n + b_n)/\sqrt{V_n} = o(1)$ (the same assumptions as Theorem 3.1). Then, $\hat{V}_n/V_{n,P} \xrightarrow{P} 1$.

*Proof.* Denote $\|f\|_{n,2} := (\mathbb{E}_n\|f_i\|_2^2)^{1/2}$. Recall $\hat{V}_n = \mathrm{Var}_n(\hat{S}_{ni})$. We first establish $\mathbb{E}_n\|\hat{S}_{ni} - S_{ni}\|_2/\sqrt{V}_n = o_p(1)$, then show $|\mathbb{E}_n S_{ni}^2/V_n - 1| = o_p(1)$.

$$\hat{S}_{ni} - S_{ni} = l_n^\top(\hat{H}_n^{-1} - H_n^{-1})\psi_{ni} + l_n^\top \hat{H}_n^{-1}(\hat{\psi}_{ni} - \psi_{ni}) + l_n^\top \hat{H}_n^{-1}\{\hat{\psi}_{ni}(\hat{\beta}_n) - \hat{\psi}(\beta_{P,n})\}.$$

The first term is by the Cauchy-Schwarz inequality and Equation (16),

$$\|l_n^\top(\hat{H}_n^{-1} - H_n^{-1})\psi_{ni}\|_{n,2} \leq \|l_n\|_2\|\hat{H}_n^{-1}\|_{op}\|\hat{H}_n - H_n\|_{op}\|H_n^{-1}\psi_{ni}\|_{n,2} = O_p(\|l_n\|_2 a_n) \quad (19)$$

Similarly, $\|l_n^\top \hat{H}_n^{-1}(\hat{\psi}_{ni} - \psi_{ni})\|_{n,2} = O_p(\|l_n\|_2 b_n)$ from the second part of Equation (18) and the Cauchy-Schwarz inequality. To bound the last term,

$$\begin{aligned}\|\hat{\psi}_{ni}(\hat{\beta}_n) - \hat{\psi}(\beta_{P,n})\|_{n,2}^2 &= \left\|\frac{\hat{\mu}_{qi}q_i^\top + (q_i - \hat{\mu}_{qi}\hat{\mu}_{qi})^\top}{\sigma_i^2}(\hat{\beta}_n - \beta_{P,n})\right\|_{n,2}^2 \\ &:= \frac{1}{n}\sum_i \left\|\hat{H}_{ni}(\hat{\beta}_n - \beta_{P,n})\right\|_2^2\end{aligned}$$

For any $v \in \mathbf{R}^{k_n}$, by Assumption 3.4(iii), the Cauchy-Schwarz inequality,

$$\begin{aligned}\|\hat{H}_{ni}v\|_2^2 &= \|\frac{1}{\sigma_i^2}\big(\mu_{qi}q^\top v + (q_i - \mu_{qi})\mu_q^\top v + e_\mu(q_i - \mu_{qi})^\top v + (q_i - \mu_{qi})e_\mu^\top v - e_\mu e_\mu^\top v\big)\|_2^2 \\ &\precsim \|\mu_{qi}q^\top v\|_2^2 + \|(q_i - \mu_{qi})\mu_q^\top v\|_2^2 + \|e_\mu(q_i - \mu_{qi})^\top v\|_2^2 + \|(q_i - \mu_{qi})e_\mu^\top v\|_2^2 + \|e_\mu e_\mu^\top\|_2^2 \\ &\leq \xi_n^2|q_i^\top v|^2 + 4\xi_n^2|\mu_q^\top v|^2 + \|e_\mu\|_2^2|(q_i - \mu_{qi})^\top v|^2 + 4\xi_n^2|e_\mu^\top v|^2 + \|e_\mu\|_2^2|e_\mu^\top v|^2 \\ \mathbb{E}_n\|\hat{H}_{ni}v\|_2^2 &\precsim \xi_n^2 v^\top \mathbb{E}_n[q_i q_i^\top]v + 4\xi_n^2 v^\top \mathbb{E}_n[\mu_{qi}\mu_{qi}^\top]v + O_p(\xi_n^2 r_{\mu,n}^2\|v\|_2^2) + O_p(\xi_n^2 r_{\mu,n}^2\|v\|_2^2) \\ &\quad + \mathbb{E}_n\|e_\mu\|_2^4\|v\|_2^2.\end{aligned}$$

The first two terms are $O_p(\xi_n^2\|v\|_2^2)$ because $\mathbb{E}_n[q_i q_i^\top]$ and $\mathbb{E}_n[\mu_{qi}\mu_{qi}^\top]$ have a bounded operator norm due to Lemma 6.2 of Belloni et al. (2015) as in Lemma B.1. To

bound the last term, since $(\mathbb{E}_n\|e_\mu\|_2^2)^2 = n^{-1}\mathbb{E}_n\|e_\mu\|_2^4 + 2n^{-2}\sum_{i>j}\|e_{\mu,i}\|_2^2\|e_{\mu,j}\|_2^2 \geq n^{-1}\mathbb{E}_n\|e_\mu\|_2^4$, multiplying $n$ both sides yields

$$\frac{1}{n}\sum_i \|e_\mu\|_2^4 \leq n(\mathbb{E}_n\|e_\mu\|_2^2)^2 = O_p(nr_{\mu,n}^4).$$

The above bound on $\|\hat{H}_{ni}v\|_2^2$ holds for any $v \in \mathbf{R}^{k_n}$. Therefore, $\|\hat{\psi}_{ni}(\hat{\beta}_n) - \hat{\psi}_{ni}(\beta_{P,n})\|_{n,2} = O_p(\{\xi_n + \xi_n r_{\mu,n} + \sqrt{n}r_{\mu,n}^2\})\|\hat{\beta}_n - \beta_{P,n}\|_2$. From Equations (14), (15), and Lemma 3.1, $\|\hat{\beta}_n - \beta_{P,n}\|_2 = O_p(n^{-1/2}\{\tau_n^2\sqrt{k_n + \xi_n^2 r_{w,n}^2} + a_n + b_n\})$. Combining all three terms,

$$\begin{aligned}\|l_n^\top \hat{H}_n^{-1}\{\hat{\psi}_{ni}(\hat{\beta}_n) - \hat{\psi}_{ni}(\beta_{P,n})\}\|_{n,2} &= O_p\Big(\|l_n\|_2\tau_n^2(\frac{\xi_n}{\sqrt{n}} + \frac{\xi_n r_{\mu,n}}{\sqrt{n}} + r_{\mu,n}^2)(\tau_n\sqrt{k_n} + \tau_n^2\xi_n r_{w,n} + a_n + b_n)\Big) \\ &= O_p(\|l_n\|_2(a_n + b_n)),\end{aligned}$$

where the last step uses the assumption that $\tau_n^2 r_{H,n} = o(1)$.

Finally, under the assumption that $\|l_n\|_2(a_n+b_n)/\sqrt{V}_n = o_p(1)$, $\|\hat{S}_{ni} - S_{ni}\|_{n,2}/\sqrt{V_n} = o_p(1)$.

To conclude the proof, under Assumption 3.6(ii), apply, for example, Proposition 1 of Portier and Segers (2019) with $Y_{ni} = \frac{S_{ni}^2}{V_n}$. Then, $\mathbb{E}_n Y_{ni} \xrightarrow{p} 1$, hence, $|S_{ni}^2/V_n - 1| \xrightarrow{p} 0$ as desiered.

□

## B.2 Proofs for Appendix A

*Proof of Lemma A.1 (Gaussian coupling).* Let $\mathbb{G}_n\psi_{n,P}(O_i) := \frac{1}{\sqrt{n}}\sum_i\{\psi_{n,P}(O_i) - \mathbb{E}_P[\psi_{n,P}(O)]\}$. From the interior solution assumption, $\mathbb{E}_P[\psi_{n,P}] = 0$. Multiplying $D_{n,P}$ to the linear expansion in Lemma 3.1,

$$\|\sqrt{n}D_{n,P}(\hat{\beta}_n - \beta_{P,n}) - W_{n,P}^D\|_2 \leq \| - D_{n,P}H_{n,P}^{-1}\mathbb{G}_n\psi_{n,P} - W_{n,P}^D\|_2 + \|D_{n,P}R_{n,P}\|_2.$$

We may apply Theorem 10.10 of Pollard (2002) to the empirical process term and show the remainder is $o(c_n)$. Choose $\epsilon_n \to 0$ sufficiently slowly so that $\frac{k_n}{\epsilon_n^3 c_n^3}\sup_P \Delta_{D,P} \to$

0. Applying the theorem with $\delta = \epsilon_n c_n$ and $B = \frac{k_n \Delta_{D,P}}{\epsilon_n^3 c_n^3}$,

$$P(\|D_{n,P} H_{n,P}^{-1} \mathbb{G}_n \psi_{n,P} - W_{n,P}^D\|_2 > 3\epsilon_n c_n) \leq C_0 B\{1 + \frac{|\log(1/B)|}{k_n}\} \to 0$$

for some universal constant $C_0$ uniformly over $P \in \mathcal{P}$. Hence, for any fixed $\eta > 0$, whenever $n$ is sufficiently large,

$$P(\|D_{n,P} H_{n,P}^{-1} \mathbb{G}_n \psi_{n,P} - W_{n,P}^D\|_2 > \eta c_n) \leq P(\|D_{n,P} H_{n,P}^{-1} \mathbb{G}_n \psi_{n,P} - W_{n,P}^D\|_2 > 3\epsilon_n c_n) \to 0.$$

Hence, $\|D_{n,P} H_{n,P}^{-1} \mathbb{G}_n \psi_{n,P} - W_{n,P}^D\|_2 = o_P(c_n)$ uniformly over $P$. The remainder term is $o_P(c_n)$ uniformly over $P$ by Assumption A.1(iv). □

# C Verification of Assumption 3.3

**Lemma C.1** (Verification of Assumption 3.3 for KLS)**.** Suppose the following conditions hold. (i) $Z \in [0,1]^{dz}$ and the marginal density is bounded away from 0 uniformly over $P$. (ii) $\|\mu_{n,P}(z) - \mu_{n,P}(z')\|_2 \leq L_{\mu,n}\|z - z'\|_2$, for any $z, z' \in \mathcal{Z}_P$ for every $P \in \mathcal{P}$ with the Lipschitz constant independent of $P$. (iii) There exists constatns $C_c, C_q < \infty$ independent of $n$, $P$, and a deterministic sequence $\xi_n$ independent of $P$, such that $\|\mathrm{Var}_P(c_n(O)|Z)\|_{P,\infty} \leq C_c^2$ and $\|q_n(O)\|_2 \leq C_q \xi_n$ $P$-a.s.. (iv) Subsample size $s_n \to \infty$, $s_n/n \to 0$, $\kappa$ is fixed, and the number of subsampling $B_n \gtrsim n/s_n$.

Then, for an evaluation draw $Z \sim P_Z$ independent of $\mathcal{T}_\ell$,

$$\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2 | \mathcal{T}_\ell] = O_P(L_{\mu,n}^2 s_n^{-2/d_z} + (1 + \xi_n^2)\frac{s_n}{n}).$$

Condition (ii) depends on the choice of the sieve dictionary. The Lipschitz constant $L_{\mu,n}$ may grow with $n$, as the number of the sieve dictionary grows. Condition (iii) is direct from the envelop condition in the main text.

*Proof of Lemma C.1.* Fix the fold $\ell$, the evaluation index set $I_\ell$, the training index set $J_\ell$, and an evaluation point $z$. Let $n_J := |J| = n - |I|$. Let $A$ denote all the

auxiliary learner randomness (such as subsampling, tie-breaking randomness), which is independent of the observations. Define the sigma field $\mathcal{D} = \sigma(\{Z_j\}_{j\in J_\ell}, A)$. By construction, the weights depend only on the point $z$, the conditioning variables $\{Z_j, j \in J\}$, and the auxiliary randomness $A$. Hence, the weights are $\mathcal{D}$-measurable and considered as fixed real numbers conditional on $\mathcal{D}$. Decompose the prediction error as

$$\hat{\mu}_n^{(-\ell)}(z) - \mu_{n,P}(z) = \sum_{j\in J_\ell} \alpha_j^{(-\ell)}(z)\{\begin{pmatrix} c_n(O_j) \\ q_n(O_j) \end{pmatrix} - \mu_{n,P}(Z_j)\} + \sum_{j\in J_\ell} \alpha_j^{(-\ell)}(z)\{\mu_{n,P}(Z_j) - \mu_{n,P}(z)\}.$$

Conditional on $\mathcal{D}$, the first term has mean 0, and the second term is fixed. Hence, the cross-product between the two terms has mean 0. Then,

$$\begin{aligned} \mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(z) - \mu_{n,P}(z)\|_2^2|\mathcal{D}] &= \|\sum_{j\in J_\ell} \alpha_j^{(-\ell)}(z)\{\mu_{n,P}(Z_j) - \mu_{n,P}(z)\}\|_2^2 \\ &\quad + \sum_{j\in J_\ell} \alpha_j^{(-\ell)}(z)^2 \mathbb{E}_P[\|(c_n(O_j), q_n(O_j))^\top - \mu_{n,P}(Z_j)\|_2^2|Z_j]. \end{aligned}$$

By the Jensen inequality and the Lipschitzness assumption,

$$\begin{aligned} \|\sum_{j\in J_\ell} \alpha_j^{(-\ell)}(z)\{\mu_{n,P}(Z_j) - \mu_{n,P}(z)\}\|_2^2 &\leq \sum_{j\in J_\ell} \alpha_j^{(-\ell)}(z)\|\mu_{n,P}(Z_j) - \mu_{n,P}(z)\|_2^2 \\ &\leq L_{\mu,n}^2 \sum_{j\in J_\ell} \alpha_j^{(-\ell)}(z)\|Z_j - z\|_2^2 \\ &= \frac{L_{\mu,n}^2}{B_n} \sum_b \sum_{j\in J_\ell} \alpha_{b,j}^{(-\ell)}(z)\|Z_j - z\|_2^2 \\ &\leq \frac{L_{\mu,n}^2}{B_n} \sum_b R_\kappa(z, S_b)^2. \end{aligned}$$

where $R_\kappa(z, S_b)$ denotes the distance from $z$ to the $\kappa$-th nearest neighbor in subsample

$S_b$. Meanwhile, by the bounded conditional second moment assumption,

$$\begin{aligned}\mathbb{E}_P[\|(c_n(O_j), q_n(O_j))^\top - \mu_{n,P}(Z_j)\|_2^2 | Z_j] &= \mathrm{Var}_P(c_n(O_j)|Z_j) + \mathbb{E}_P[\|q_n(O_j)\|_2^2|Z_j] - \|\mu_q(Z_j)\|_2^2 \\ &\leq C_c^2 + C_q^2 \xi_n^2.\end{aligned}$$

Combining the bounds and taking expectation over $\mathcal{D}$,

$$\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(z) - \mu_{n,P}(z)\|_2^2] \leq L_{\mu,n}^2 \mathbb{E}_P[R_\kappa(z,S)^2] + (C_c^2 + C_q^2\xi_n^2)\mathbb{E}_P[\sum_{j \in J_\ell} \alpha_j^{(-\ell)}(z)^2].$$

Now we bound the two expectation terms on the right-hand side. Under the condition (i), for small $t \in (0, \frac{1}{2})$,

$$P_Z\{B(z,t)\} = \int_{B(z,t)\cap[0,1]^{d_z}} f_{P,Z}(u)du \geq \underline{f} 2^{-d_z} v_{d_z} t^{d_z}$$

where $v_{dz}$ is the volume of $d_z$-dimensional unit ball. Then, their Lemma 2 and the tail bound in Appendix H.2 imply that there exist positive constants $c$ and $C_0$ independent of $P$, $z$, $\ell$, such that, for sufficiently large $n$, for $t \in [C_0 s_n^{-1/d_z}, r_0]$,

$$P\{R_\kappa(z,S) > t\} \leq 2\exp(-cs_n t^{d_z}).$$

Following their integration argument in their Appendix H.2 on Page 27[8] with the identity $\mathbb{E}[X^2] = \int_0^\infty 2tPr(X > t)dt$ and $t_n = C_0 s_n^{-1/d_z}$, we obtain $\mathbb{E}_P[R_\kappa(z,S)^2] \leq C_R s_n^{-2/d_z}$ for some $C_R > 0$ uniformly over $P$ and $P$-almost every $z$. We show the mean of the sum of squared weights is $O(\frac{1}{\kappa}(\frac{1}{B_n} + \frac{(B_n-1)s_n}{B_n n_J}))$. Hence, for sufficiently

[8]where they obtain the first moment of $R_\kappa$ in their analysis, while we need the second moment.

large $B_n$, it is $O(\frac{s_n}{\kappa n_J})$.

$$\begin{aligned}\mathbb{E}_P[\sum_{j\in J_\ell}(\alpha_j^{(-\ell)}(z))^2|Z_j, j\in J_\ell] &= \frac{1}{B_n^2}\sum_{b=1}^{B_n}\sum_{j\in J_\ell}\mathbb{E}_P[(\alpha_{b,j}^{(-\ell)}(z))^2|Z_j, j\in J_\ell] \\ &\quad +\frac{1}{B_n^2}\sum_{b\neq b'}^{B_n}\sum_{j\in J_\ell}\mathbb{E}_P[\alpha_{b,j}^{(-\ell)}(z)\alpha_{b',j}^{(-\ell)}(z)|Z_j, j\in J_\ell] \\ &= \frac{1}{\kappa B_n}+\frac{B_n(B_n-1)}{B_n^2}\sum_{j\in J_\ell}\mathbb{E}_P[\alpha_{b,j}^{(-\ell)}(z)|Z_j, j\in J_\ell]^2\end{aligned}$$

because $b$ and $b'$ are independent, and $\alpha_{b,j}^{(-\ell)}(z) = \frac{1}{\kappa}$ if $j$ is in $\kappa$-nearest neighbor in subsample $S_b$, and 0 otherwise. Meanwhile, $\mathbb{E}_P[\alpha_{b,j}^{(-\ell)}(z)|Z_j, j\in J_\ell]$ has natural worst-case and best-case bounds 0 and $s_n/\kappa n_J$, and sums up to 1 by taking the conditional expectation to the identity $\sum_{j\in J_\ell}\alpha_{b,j}^{(-\ell)}(z) = 1$. Hence,

$$\begin{aligned}\sum_{j\in J_\ell}\mathbb{E}_P[\alpha_{b,j}^{(-\ell)}(z)|Z_j, j\in J_\ell]^2 &\leq (\max_{j\in J_\ell}\mathbb{E}_P[\alpha_{b,j}^{(-\ell)}(z)|Z_j, j\in J_\ell])\sum_{j\in J_\ell}\mathbb{E}_P[\alpha_{b,j}^{(-\ell)}(z)|Z_j, j\in J_\ell] \\ &\leq \frac{s_n}{\kappa n_J}.\end{aligned}$$

Therefore, for a fixed point $z$ and sufficiently large $B_n \geq n_J/s_n$, we obtain the explicit finite sample bound

$$\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(z) - \mu_{n,P}(z)\|_2^2] \leq (C_c^2 + C_q^2\xi_n^2)\frac{s_n}{\kappa n_J} + C_R L_{\mu,n}^2 s_n^{-2/d_z} = O(r_{\mu,n}^2),$$

where $r_{\mu,n}^2 = (C_c^2 + C_q^2\xi_n^2)\frac{s_n}{\kappa n} + L_{\mu,n}^2 s_n^{-2/d_z}$. We use $n \asymp n_J$ in the last equality. Note that the above bound holds for $z$-a.s., with a constant independent of $P$, $z$, and the fold $\ell$.

Finally, to obtain the desired result in Assumption 3.3, let $Z \sim P_Z$ be an inde-

pendent evaluation draw. Then,

$$\begin{aligned}\mathbb{E}_P\Big[\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(Z)-\mu_{n,P}(Z)\|_2^2|\mathcal{T}_\ell]\Big] &= \mathbb{E}_P[\int_Z \|\hat{\mu}_n^{(-\ell)}(Z)-\mu_{n,P}(Z)\|_2^2 dP_Z] \\ &= \int_Z \mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(Z)-\mu_{n,P}(Z)\|_2^2] dP_Z \\ &\leq Cr_{\mu,n}^2,\end{aligned}$$

for some fixed $C > 0$. By the Markov inequality $\mathbb{E}_P[\|\hat{\mu}_n^{(-\ell)}(Z)-\mu_{n,P}(Z)\|_2^2|\mathcal{T}_\ell] = O_p(r_{\mu,n}^2)$ uniformly over $P \in \mathcal{P}$. □

**Lemma C.2** (Verification of Assumption 3.3 for ATW (GRF))**.** Suppose the observations are i.i.d., the training folds satisfy $|\mathcal{T}_\ell| \asymp n$, and the following conditions hold uniformly over $P \in \mathcal{P}$. (i) $Z \in [0,1]^{d_z}$ for a fixed $d_z$ and the density of $Z$ is bounded above and away from zero by constants independent of $P$. (ii) $\|\mu_{n,P}(z)-\mu_{n,P}(z')\|_2 \leq L_{\mu,n}\|z-z'\|_2$ for some deterministic sequence $L_{\mu,n}$. (iii) There exist constants $C_c, C_q < \infty$, independent of $n, P$, such that $\operatorname{ess\,sup}_z \mathbb{E}_P[c_n(O)^2 \mid Z = z] \leq C_c^2, \|q_n(O)\|_2 \leq C_q\xi_n \quad P$-a.s. (iii) The forest is honest and regular in the sense of Specification 1 of Athey, Tibshirani, and Wager (2019), with a fixed leaf size. (iv) Subsample size $s_n \to \infty$, the number of trees $B_n \gtrsim n/s_n$.

Then, for $Z \sim P_Z$ independent of $\mathcal{T}_\ell$,

$$\mathbb{E}_P\left[\|\hat{\mu}_n^{(-\ell)}(Z)-\mu_{n,P}(Z)\|_2^2|\mathcal{T}_\ell\right] = O_P\Big(L_{\mu,n}^2 s_n^{-2\gamma} + (1+\xi_n^2)\frac{s_n}{n}\Big)$$

uniformly over $P \in \mathcal{P}$. Here, $s_n$ is a tuning parameter and $\gamma$ is a constant[9] decided from some other tuning parameters and $d_z$.

*Proof of Lemma C.2.* Fix a fold $\ell$ and an evaluation point $z$. Let the training index set be $J_\ell$ and write $n_J = |J_\ell|$. Let $J_{\ell b} \subset J_\ell$ be the subsample for $b = 1, ..., B_n$. For

[9] $\gamma := \frac{\pi}{2d_z}\frac{\log((1-w)^{-1})}{\log(w^{-1})}$ where $\pi$ is a constant from 'each coordinate is selected for splitting with probability at least $\pi/d_z$', $w$ is a minimum split fraction. Stronger balance $w$ might exclude useful split – learn data less, but prevent worst outcome. It affects the bias term (larger $w$ means larger $\gamma$. The default choice of 'mtry' in GRF package does not guarantee a specific $\pi$. One may force it by using a different feature selection rule, e.g., uniformly choose one among $d_z$ instruments, so that $\pi = 1$. We stick to the default 'mtry' in our implementation.

honest split, let $J_{\ell b} = J_{\ell b1} \cup J_{\ell b2}$, where $J_{\ell b1}$ is the evaluation split and $J_{\ell b2}$ is the partitioning split. Let $A$ contain all auxiliary randomness used for subsampling, honest splitting, feature selection, and tie-breaking. $A$ is independent of the observations. Define a sigma field $\mathcal{D}_\ell := \sigma(\{Z_j, O_j : j \in J_\ell\}, A)$. Let index $b$ denote an individual tree. Write the tree prediction

$$T_b(z) = \sum_{i \in J_\ell} \alpha_{b,i}(z) \begin{pmatrix} c_n(O_i) \\ q_n(O_i) \end{pmatrix}, \qquad \hat{\mu}_n^{(-\ell)}(z) = \frac{1}{B_n} \sum_{b=1}^{B_n} T_b(z),$$

where $\alpha_{b,i}(z) \geq 0$ and $\sum_i \alpha_{b,i}(z) = 1$. See Athey, Tibshirani, and Wager (2019) for the exact definition. Define the finite and infinite forest

$$\hat{\mu}_n^{(-\ell)}(z) = \frac{1}{B_n} \sum_{b=1}^{B_n} T_b(z), \qquad \hat{\mu}_{\infty,n}^{(-\ell)}(z) := \mathbb{E}_A[T_b(z) | O_j, j \in J_\ell].$$

where the expectation is taken over tree randomness $A_b$. Note $\mathbb{E}_{P,A}[\hat{\mu}_n^{(-\ell)}(z)] = \mathbb{E}_P[\hat{\mu}_\infty^{(-\ell)}(z)]$.

Decomposing the mean squared error,

$$\begin{aligned}
&\mathbb{E}_{P,A} \|\hat{\mu}_n^{(-\ell)}(z) - \mu_{n,P}(z)\|_2^2 \\
&= \mathbb{E}_{P,A} \left\| \{\mathbb{E}_{P,A}[\hat{\mu}_n^{(-\ell)}(z)] - \mu_{n,P}(z)\} + \{\hat{\mu}_{\infty,n}^{(-\ell)}(z) - \mathbb{E}_P[\hat{\mu}_{\infty,n}^{(-\ell)}(z)]\} + \{\hat{\mu}_n^{(-\ell)}(z) - \hat{\mu}_{\infty,n}^{(-\ell)}(z)\} \right\|_2^2 \\
&= \|\mathbb{E}_{P,A}[\hat{\mu}_n^{(-\ell)}(z)] - \mu_{n,P}(z)\|_2^2 + \mathbb{E}_P \|\hat{\mu}_{\infty,n}^{(-\ell)}(z) - \mathbb{E}_P[\hat{\mu}_{\infty,n}^{(-\ell)}(z)]\|_2^2 + \mathbb{E}_{P,A} \|\hat{\mu}_n^{(-\ell)}(z) - \hat{\mu}_{\infty,n}^{(-\ell)}(z)\|_2^2.
\end{aligned}$$

The first term in the decomposition is deterministic, the second term has mean 0, and the last term has mean 0 conditionally on $\{(Z_i, O_i) : i \in J_\ell\}$. Hence, the cross product terms are 0 by iterative expectation, and we obtain the last equality.

The first term is $\|\mathbb{E}_P[\hat{\mu}_{\infty,n}^{(-\ell)}(z)] - \mu_{n,P}(z)\|_2^2 \leq C L_{\mu,n}^2 s_n^{-2\gamma}$ by the Lipschitz condition and the localization bound in Equation (26) of ATW18. The second term satisfies

$$\mathbb{E}_P \|\hat{\mu}_{\infty,n}^{(-\ell)}(z) - \mathbb{E}_P[\hat{\mu}_{\infty,n}^{(-\ell)}(z)]\|_2^2 = \operatorname{tr}(\operatorname{Var}_P(\hat{\mu}_{\infty,n}^{(-\ell)}(z))) \leq \frac{s_n}{n_J}(C_c^2 + C_q^2 \xi_n^2),$$

where the inequality follows by applying the first ineqaulity in Equation (27) of

ATW18 to our conditional mean score $\psi_{\mu_{n,P}(z)}(O_i) = (c_n, q_n^\top)^\top - \mu_{n,P}(z)$, Jensen's inequality, honesty, and the condition (ii) in Lemma C.2. For the last term, by LIE conditioning on $\{Z_i, O_i : i \in J_\ell\}$, conditional independency and mean 0 of the trees,

$$\begin{aligned}\mathbb{E}_{P,A}\big[\|\hat{\mu}_n^{(-\ell)}(z) - \hat{\mu}_{\infty,n}^{(-\ell)}(z)\|_2^2\big] &= \frac{1}{B_n}\mathbb{E}_{P,A}\big[\|T_1(z) - \hat{\mu}_{\infty,n}^{(-\ell)}(z)\|_2^2\big] \\ &\leq \frac{1}{B_n}\mathbb{E}_{P,A}\|T_1(z)\|_2^2 \leq \frac{C_c^2 + C_q^2\xi_n^2}{B_n}.\end{aligned}$$

The last inequality follows from Jensen's inequality, LIE conditioning on $\{A_1, (Z_i, O_i) : i \in J_{\ell 11}\}$, honesty, condition (ii), and $\sum_i \alpha_{1,i}(z) = 1$.

Substituting the three bounds into the mean squared error decomposition and taking $B_n \geq n_J/s_n$, we obtain

$$\mathbb{E}_{P,A}\|\hat{\mu}_n^{(-\ell)}(z) - \mu_{n,P}(z)\|_2^2 \leq 2(C_c^2 + C_q^2\xi_n^2)\frac{s_n}{n_J} + CL_{\mu,n}^2 s_n^{-2\gamma}.$$

The constants are independent of $P, z, \ell$.

Finally,

$$\begin{aligned}\mathbb{E}_{P,A}\left[\mathbb{E}_P\left[\|\hat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2 \mid \mathcal{D}_\ell\right]\right] &= \int \mathbb{E}_{P,A}\|\hat{\mu}_n^{(-\ell)}(z) - \mu_{n,P}(z)\|_2^2 \, dP_Z(z) \\ &= O_P(L_{\mu,n}^2 s_n^{-2\gamma} + (1+\xi_n^2)\frac{s_n}{n}),\end{aligned}$$

where the last equality uses $n_J \asymp n$. The Markov's inequality yields

$$\mathbb{E}_P\big[\|\hat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2 \mid \mathcal{D}_\ell\big] = O_P(L_{\mu,n}^2 s_n^{-2\gamma} + (1+\xi_n^2)\frac{s_n}{n})$$

uniformly over $P \in \mathcal{P}$. □

**Lemma C.3** (Verification of the first-stage assumption for conditional diffusion)**.** Write $O = (V, Z)$, where $V \in \mathbb{R}^{d_V}$ and $d_V, d_z$ are fixed, and write $c(O) = c(V, Z)$ and $q_n(O) = q_n(V, Z)$. Suppose the following additional conditions hold, with constants independent of $P$, $n$, and the fold.

Suppose Assumptions 3.1(i)(ii),(iv)–(vi) and 3.2(i)(ii) hold. Moreover, suppose

the following conditions hold. (i) Let $Z \in [0,1]^{d_z}$. Let $\varphi_\lambda$ be the density of $N(0, \lambda^{-1} I_{d_V})$ for a fixed $\lambda > 0$. The conditional density of $V$ given $Z = z$ satisfies $f_{0,P}(v \mid z) = \varphi_\lambda(v) g_P(v, z)$, where $g_P$ is bounded above and away from zero and belongs to a $s$-Hölder ball on $\mathbb{R}^{d_V} \times [0,1]^{d_z}$. (ii) The architecture of the neural network and tuning parameters satisfy the specification in Theorems 3.4 and 4.2 of Fu et al. (2024). (iii) The estimated conditional distribution satisfies $\operatorname{ess\,sup}_z \int c(v,z)^2 \, d\hat{P}_n^{(-\ell)}(v \mid z) = O_P(1)$ uniformly over $P \in \mathbf{P}$.

Define

$$\widehat{\mu}_n^{(-\ell)}(z) = \int \begin{pmatrix} c(v,z) \\ q_n(v,z) \end{pmatrix} d\widehat{P}_n^{(-\ell)}(v \mid z).$$

Then, for $Z \sim P_Z$ independent of $\mathcal{T}_\ell$,

$$\mathbb{E}_P\big[\|\widehat{\mu}_n^{(-\ell)}(Z) - \mu_{n,P}(Z)\|_2^2 \mid \mathcal{T}_\ell\big] = O_P\big((1 + \xi_n^2)\varepsilon_n^2\big) \tag{20}$$

uniformly over $P$ where $\varepsilon_n^2 = n^{-2s/(2s+d_V+d_z)} (\log n)^{\max\{19, s+2\}}$.

In Condition (i), the support of $Z$ is for convenience and can be relaxed to unbounded distributions with small tails (Fu et al., 2024, Appendix C). Condition (i) requires the conditional density to be a smooth, uniformly bounded positive multiple of a fixed Gaussian density. It allows non-Gaussian conditional distributions but imposes restrictive support and tail conditions.